S5K3B6SX

Automotive 1/2.4" 3MP CMOS Image Sensor with On-Chip HDR & LFM

Revision 0.01

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Data Sheet (Preliminary)

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Chip Handling Guide

Precaution against Electrostatic Discharge

When using semiconductor devices, ensure that the environment is protected against static electricity:

1. Wear antistatic clothes and use earth band.
2. All objects that are in direct contact with devices must be made up of materials that do not produce static electricity.
3. Ensure that the equipment and work table are earthed.
4. Use ionizer to remove electron charge.

Contamination

Do not use semiconductor products in an environment exposed to dust or dirt adhesion.

Temperature/Humidity

Semiconductor devices are sensitive to:

- Environment
- Temperature
- Humidity

High temperature or humidity deteriorates the characteristics of semiconductor devices. Therefore, do not store or use semiconductor devices in such conditions.

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Mechanical Shock

Do not to apply excessive mechanical shock or force on semiconductor devices.

Chemical

Do not expose semiconductor devices to chemicals because exposure to chemicals leads to reactions that deteriorate the characteristics of the devices.

Light Protection

In non- Epoxy Molding Compound (EMC) package, do not expose semiconductor IC to bright light. Exposure to bright light causes malfunctioning of the devices. However, a few special products that utilize light or with security functions are exempted from this guide.

Radioactive, Cosmic and X-ray

Radioactive substances, cosmic ray, or X-ray may influence semiconductor devices. These substances or rays may cause a soft error during a device operation. Therefore, ensure to shield the semiconductor devices under environment that may be exposed to radioactive substances, cosmic ray, or X-ray.

EMS (Electromagnetic Susceptibility)

Strong electromagnetic wave or magnetic field may affect the characteristic of semiconductor devices during the operation under insufficient PCB circuit design for Electromagnetic Susceptibility (EMS).

Revision History

Revision No.	Date	Description	Author(s)
0.00	June 11, 2020	Initial draft	CE Kang
0.01	April 28, 2021	Modify the contents of Fig4 (90->89)	Sungho Suh

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List of Conventions

Register RW Access Type Conventions

Type	Definition	Description
R	Read Only	The application has permission to read the register field. Writes to read-only fields have no effect.
W	Write Only	The application has permission to write in the register field.
RW	Read/Write	The application has permission to read and write in the register field. The application sets this field by writing 1'b1 and clears it by writing 1'b0. The register field is updated immediately.
RW1	Read/Write	The application has permission to read and write in the register field. The application sets this field by writing 1'b1 and clears it by writing 1'b0. The register field is updated after 1 frame.
RW2	Read/Write	The application has permission to read and write in the register field. The application sets this field by writing 1'b1 and clears it by writing 1'b0. The register field is updated after 2 frames.

Register Value Conventions

Expression	Description
x	Undefined bit
X	Undefined multiple bits
?	Undefined, but depends on the device or pin status
Device dependent	The value depends on the device
Pin value	The value depends on the pin status

Reset Value Conventions

Expression	Description
0	Clears the register field
1	Sets the register field
x	Don't care condition

Warning: Some bits of control registers are driven by hardware or write operation only. As a result, the indicated reset value and the read value after reset should be different.

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List of Acronyms

Acronyms	Descriptions
ADLC	Auto Dark Level Compensation
APS	Active Pixel Sensor
CDS	Correlated Double Sampling
CIS	CMOS Image Sensor
FPN	Fixed Pattern Noise
WDR	Wide Dynamic Range (High Dynamic Range)
HSYNC	Horizontal Sync. Signal
OTP	One Time Programmable
PLL	Phase-Locked Loop
TG	Timing Generator
VSYNC	Vertical Sync. Signal

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1 Product Overview

This chapter includes the following sections:

- Introduction
- Features

1.1 Introduction

S5K3B6SX is a highly integrated 3MP camera chip that includes a CMOS Image Sensor (CIS) with image correction functionality and serial transmission capability using 4-lane MIPI. It is designed for delivering the full resolution pixels of 60 frames per second (fps) with 12-bit ADC, 2-exposure, intra-scene dual conversion gain, and compressed outputs for Wide Dynamic Range (WDR) image with LED Flicker Mitigation.

S5K3B6SX is fabricated by the SAMSUNG 65 nm Back Deep Trench Isolation (BDTI) CMOS image sensor process, which is developed for imaging applications to realize a high-efficiency and low-power photo sensor. And it is also fabricated by the SAMSUNG 28nm logic process for low-power. The sensor consists of 1920 x 1536 active pixels and eight pixels on each side for color interpolation. This sensor meets 1/2.4" optical format with 3 MP resolution.

S5K3B6SX has on-chip 12-bit ADC arrays to digitize the pixel output and on-chip Correlated Double Sampling (CDS) to drastically reduce the Fixed Pattern Noise (FPN). It incorporates on-chip camera functions such as Bad Pixel Correction (BPC), exposure control, WDR reconstruction, and so on.

S5K3B6SX is programmable through an I2C serial interface and includes on-chip One-Time Programmable (OTP) Non-Volatile Memory (NVM).

S5K3B6SX is suitable for an automotive camera module with 2.8 V/1.8 V/1.05 V power supply, and the target applications are E-Mirror, Front/Rear-View camera and Surround-View-Monitoring (SVM) applications.

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1.2 Features

S5K3B6SX supports the following features:

- 3 MP sensor with 1/2.4" optics
- Unit pixel size : 3.0 μm Split PD
- Effective resolution of 1920(H) x 1536 (V) with 3MP
- Active resolution of 1936 (H) x 1552 (V) with 3MP
  - Including eight pixels in each side for color processing
- High dynamic range and high sensitivity
- Electronic rolling shutter
- Maximum frame rate
  - Compressed 12-bit 3MP 60 fps (Intra-scene*1) Dual Conversion gain with LFM)
  - Compressed 12-bit 3MP 60 fps (Intra-scene* Dual Conversion gain without LFM)
- Interfaces
  - Fine interface frequency control using additional dedicated PLL for EMI avoidance and integration flexibility
  - 2~4-lane selectable MIPI CSI-2 interface (Data rate : 4.5Gbps/lane)*2)
- Output formats
  - 24bit Combined RAW, 20/16/14/12-bit compressed combined RAW @ Intra-scene Dual conversion gain without LFM.
  - 20bit Combined RAW, 16/14/12-bit compressed combined RAW @ Intra-scene Dual conversion gain with LFM
- Control interface
  - CCI (Camera Control Interface) High speed I2C-compatible-Two-wire serial communication circuit up to 1MHz(FastMode+)
- 128 Kbit on-chip OTP memories to support defect corrections and chip ID
- Analog gain up to x16
- Digital gain up to x256
- Vertical flip and horizontal mirror mode
- Static bad pixel correction with OTP
- Statistics data output for Combined image
- Output Defect and Detection map during image data stream. (RAW8)
- Built-in test pattern generation
- Embedded temperature sensor
- Supply voltage of 2.8 V for pixel and analog, 1.8 V for I/O, and 1.05 V for digital core supply
- AEC-Q100 Grade2 Compatible

* 1) Dual conversion gain readout within single frame.

* 2) Only 4 lanes are available to support maximum frame rate operation.

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2 Pin Configuration

This chapter includes the following sections:

- Simplified I/O Symbol Diagram
- Package Dimension
- Pin Configuration
- Pin Description
- Recommended Application circuit

2.1 Simplified I/O Symbol Diagram

MCLK	I		A	VPAD
TST	I		A	VPP_CID
RSTN	I		A	VRGSL
SCE	I		A	VDRGSW
SDA	IOD		A	VMIM_PIX
SCL	IOD		A	VTG
GPIO[6:0]	IOZ		A	VNTG
VSYNC_IN	IOZ			
HSYNC_IN	IOZ		P	VDDD
I2C_SEL3	I		P	VDDIO
I2C_SEL2	I		P	VDD18_OTP
I2C_SEL1	I		P	VDD10_PLL
I2C_SEL0	I		P	M_VDD18
			P	M_VDD1
M_DNCLK_B1	OZ		P	M_VDD18_PLL
M_DPCLK_C1	OZ	S5K3B6SX	P	VDDA
M_DNDATA0_A0	OZ		P	VDDA_CDS_T
M_DPDATA0_B0	OZ		P	VDDA_CDS_B
M_DNDATA1_C0	OZ		P	VDDA28_DBR
M_DPDATA1_A1	OZ		P	VPIX_CRB_T
M_DNDATA2_A2	OZ		P	VPIX_CRB_M
M_DPDATA2_B2	OZ		P	VPIX_CRB_B
M_DNDATA3_C2	OZ		G	VSSD
M_DPDATA3	OZ		G	VSSIO
			G	VSS18_OTP
			G	VSS_PLL
			G	M_VSS
			G	VSSA
			G	VSSA28_DBR
			G	VSSA_OTA2_T
			G	VSSA_OTA2_B

Figure 1 Symbol Diagram of Simplified I/O

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NOTE: Following are the legends used in *Figure 1*:

- I: Input
- O: Output
- OZ: Tri-state output
- IOZ: Tri-state in/out
- OD: Open-drain output
- IOD: Open-drain in/out
- P: Power and ground
- A: Analog test or external capacitor node

2.2 Package Dimension

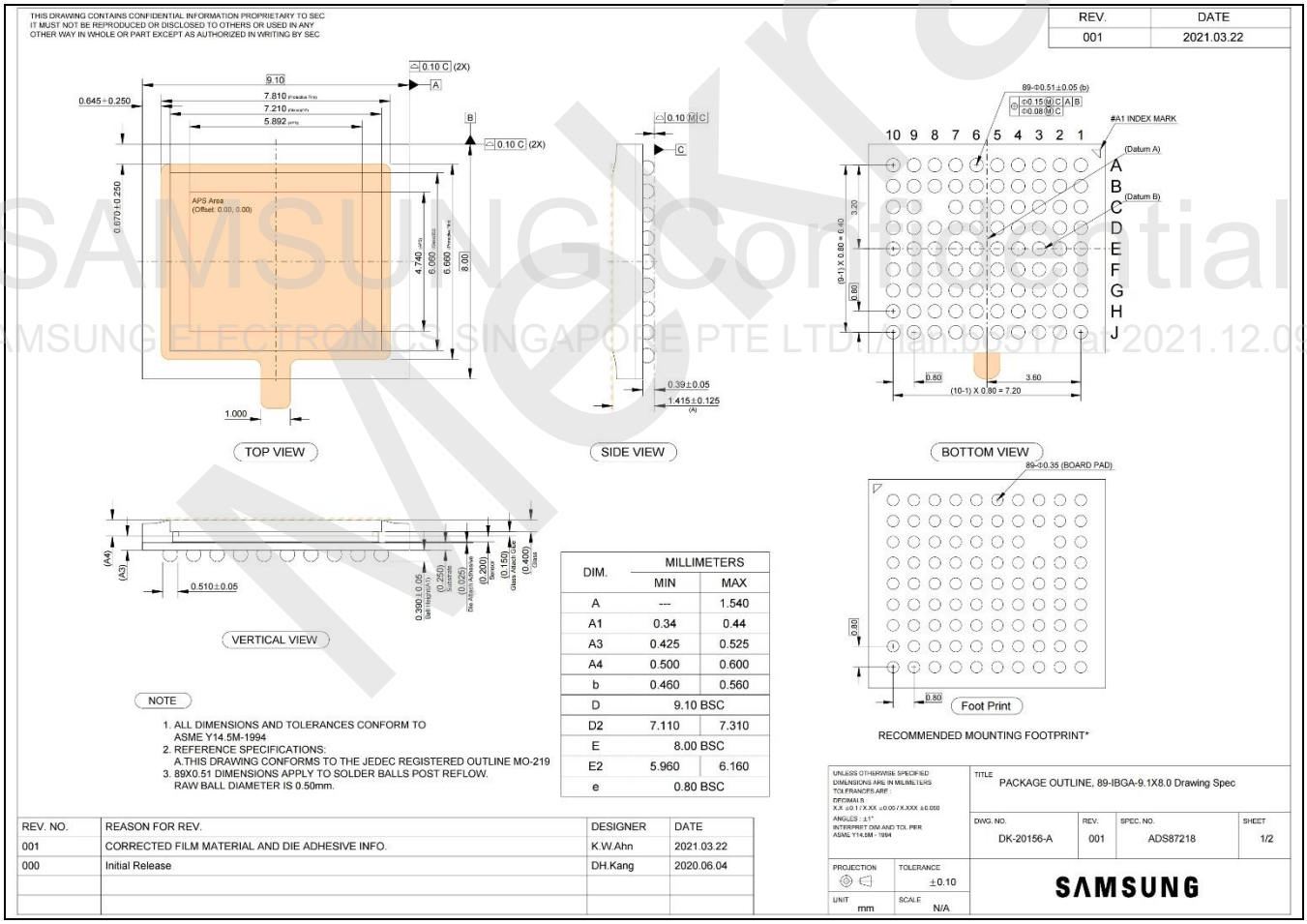
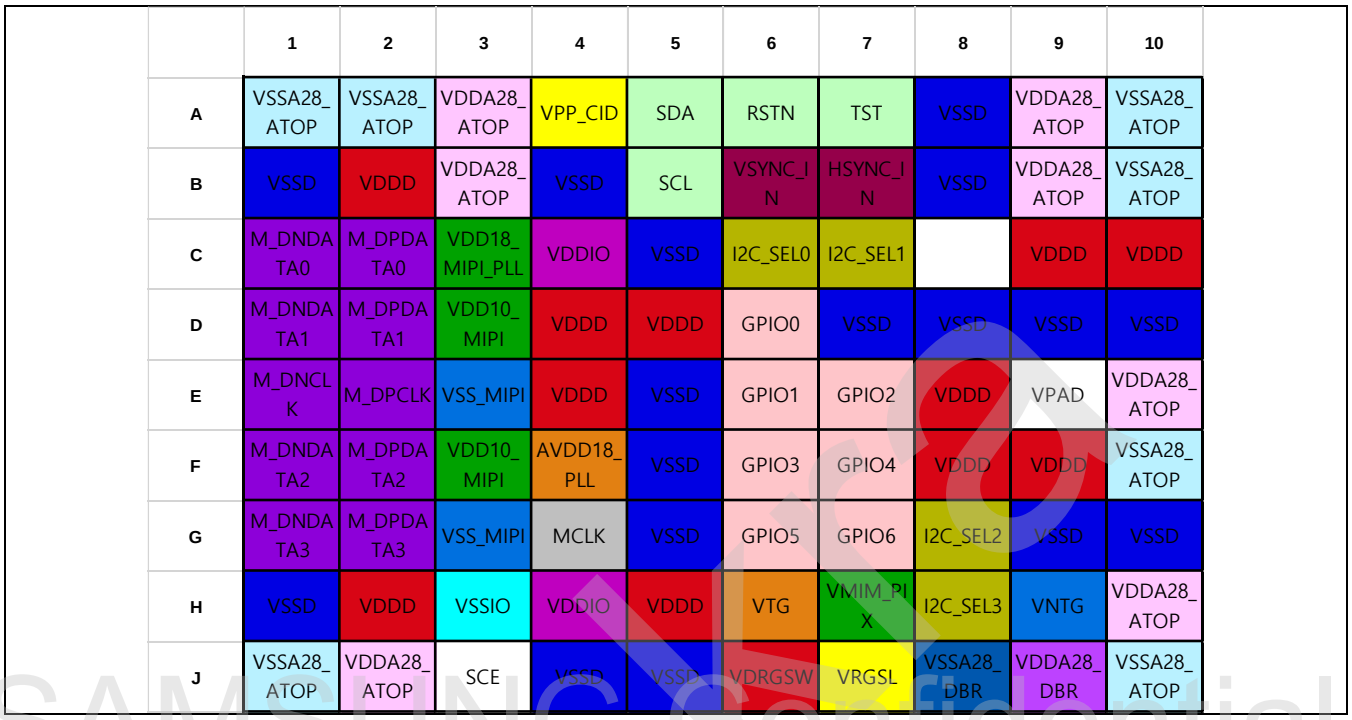


Figure 2 iBGA Package Dimension

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2.3 Pin Configuration

S5K3B6SX is assembled in iBGA package.



	1	2	3	4	5	6	7	8	9	10
A	VSSA28_ATOP	VSSA28_ATOP	VDDA28_ATOP	VPP_CID	SDA	RSTN	TST	VSSD	VDDA28_ATOP	VSSA28_ATOP
B	VSSD	VDDD	VDDA28_ATOP	VSSD	SCL	VSYNC_IN	HSYNC_IN	VSSD	VDDA28_ATOP	VSSA28_ATOP
C	M_DNDATA0	M_DPDATA0	VDD18_MIPI_PLL	VDDIO	VSSD	I2C_SEL0	I2C_SEL1		VDDD	VDDD
D	M_DNDATA1	M_DPDATA1	VDD10_MIPI	VDDD	VDDD	GPIO0	VSSD	VSSD	VSSD	VSSD
E	M_DNCLK	M_DPCLK	VSS_MIPI	VDDD	VSSD	GPIO1	GPIO2	VDDD	VPAD	VDDA28_ATOP
F	M_DNDATA2	M_DPDATA2	VDD10_MIPI	AVDD18_PLL	VSSD	GPIO3	GPIO4	VDDD	VDDD	VSSA28_ATOP
G	M_DNDATA3	M_DPDATA3	VSS_MIPI	MCLK	VSSD	GPIO5	GPIO6	I2C_SEL2	VSSD	VSSD
H	VSSD	VDDD	VSSIO	VDDIO	VDDD	VTG	VMIM_PIX	I2C_SEL3	VNTG	VDDA28_ATOP
J	VSSA28_ATOP	VDDA28_ATOP	SCE	VSSD	VSSD	VDRGSW	VRGSL	VSSA28_DBR	VDDA28_DBR	VSSA28_ATOP

Figure 3 Pin Configuration

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2.4 Pin Description

Table 1 Package Pin Descriptions

Pin No	Pin Name	Pin Type	Description
A1	VSSA28_ATOP	GROUND	Analog ground
A2	VSSA28_ATOP	GROUND	Analog ground
A3	VDDA28_ATOP	POWER	Analog power - 2.8V
A4	VPP_CID	POWER	OTP power for PGM
A5	SDA	IN/OUTPUT	I2C data
A6	RSTN	INPUT	Reset input pad
A7	TST	INPUT	Test mode control - Tie low in normal operation
A8	VSSD	POWER	Digital core ground
A9	VDDA28_ATOP	POWER	Analog power - 2.8V
A10	VSSA28_ATOP	GROUND	Analog ground
B1	VSSD	POWER	Digital core ground
B2	VDDD	POWER	Digital core power - 1.05V
B3	VDDA28_ATOP	POWER	Analog power - 2.8V

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Pin No	Pin Name	Pin Type	Description
B4	VSSD	POWER	Digital core ground
B5	SCL	INPUT	I2C clock
B6	VSYNC_IN	INPUT	Frame (Vertical) sync. input
B7	HSYNC_IN	INPUT	Line (Horizontal) sync. input
B8	VSSD	POWER	Digital core ground
B9	VDDA28_ATOP	POWER	Analog power - 2.8V
B10	VSSA28_ATOP	GROUND	Analog ground
C1	M_DNDATA0	OUTPUT	MIPI data lane 0 negative
C2	M_DPDATA0	OUTPUT	MIPI data lane 0 positive
C3	VDD18_MIPI_PLL	POWER	MIPI analog power - 1.8V
C4	VDDIO	POWER	I/O power - 1.8 V (Default)
C5	VSSD	POWER	Digital core ground
C6	I2C_SEL0	INPUT	I2C ID selection pin 0
C7	I2C_SEL1	INPUT	I2C ID selection pin 1
C8	NC	-	No connection
C9	VDDD	POWER	Digital core power - 1.05V
C10	VDDD	POWER	Digital core power - 1.05V
D1	M_DNDATA1	OUTPUT	MIPI data lane 1 negative
D2	M_DPDATA1	OUTPUT	MIPI data lane 1 positive
D3	VDD10_MIPI	POWER	MIPI digital power - 1.05V
D4	VDDD	POWER	Digital core power - 1.05V
D5	VDDD	POWER	Digital core power - 1.05V
D6	GPIO0	OUTPUT	General purpose I/O
D7	VSSD	POWER	Digital core ground
D8	VSSD	POWER	Digital core ground
D9	VSSD	POWER	Digital core ground
D10	VSSD	POWER	Digital core ground
E1	M_DNCLK	OUTPUT	MIPI clock lane negative
E2	M_DPCLK	OUTPUT	MIPI clock lane positive
E3	VSS_MIPI	POWER	MIPI ground.
E4	VDDD	POWER	Digital core power - 1.05V
E5	VSSD	POWER	Digital core ground
E6	GPIO1	OUTPUT	General purpose I/O
E7	GPIO2	OUTPUT	General purpose I/O
E8	VDDD	POWER	Digital core power - 1.05V
E9	VPAD	OUTPUT	Analog monitoring pad

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Pin No	Pin Name	Pin Type	Description
E10	VDDA28_ATOP	POWER	Analog power - 2.8V
F1	M_DNDATA2	OUTPUT	MIPI data lane 2 negative
F2	M_DPDATA2	OUTPUT	MIPI data lane 2 positive
F3	VDD10_MIPI	POWER	MIPI digital power - 1.05V
F4	AVDD18_PLL	POWER	PLL analog power - 1.8V
F5	VSSD	POWER	Digital core ground
F6	GPIO3	OUTPUT	General purpose I/O
F7	GPIO4	OUTPUT	General purpose I/O
F8	VDDD	POWER	Digital core power - 1.05V
F9	VDDD	POWER	Digital core power - 1.05V
F10	VSSA28_ATOP	GROUND	Analog ground
G1	M_DNDATA3	OUTPUT	MIPI data lane 3 negative
G2	M_DPDATA3	OUTPUT	MIPI data lane 3 positive
G3	VSS_MIPI	POWER	MIPI ground.
G4	MCLK	INPUT	External clock input
G5	VSSD	POWER	Digital core ground
G6	GPIO5	OUTPUT	General purpose I/O
G7	GPIO6	OUTPUT	General purpose I/O
G8	I2C_SEL2	INPUT	I2C ID selection pin 2
G9	VSSD	POWER	Digital core ground
G10	VSSD	POWER	Digital core ground
H1	VSSD	POWER	Digital core ground
H2	VDDD	POWER	Digital core power - 1.05V
H3	VSSIO	GROUND	I/O ground
H4	VDDIO	POWER	I/O power - 1.8V (Default)
H5	VDDD	POWER	Digital core power - 1.05V
H6	VTG	POWER	High voltage level for pixel transfer gate
H7	VMIM_PIX	POWER	MIM anode node for small PD
H8	I2C_SEL3	INPUT	I2C ID selection pin 3
H9	VNTG	POWER	Negative voltage level for pixel transfer gate
H10	VDDA28_ATOP	POWER	Analog power - 2.8V
J1	VSSA28_ATOP	POWER	Analog ground
J2	VDDA28_ATOP	POWER	Analog power - 2.8V
J3	SCE	INPUT	Clock selection (external/internal)
J4	VSSD	POWER	Digital core ground
J5	VSSD	POWER	Digital core ground

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Pin No	Pin Name	Pin Type	Description
J6	VDRGSW	POWER	Positive pixel control voltages of RG for DCG and SW for small PD.
J7	VRGSL	POWER	High voltage level for pixel reset/selection transistors
J8	VSSA28_DBR	POWER	Doubler ground
J9	VDDA28_DBR	POWER	Analog power for voltage doubler - 2.8V
J10	VSSA28_ATOP	POWER	Analog ground

2.5 Recommended Application Circuit

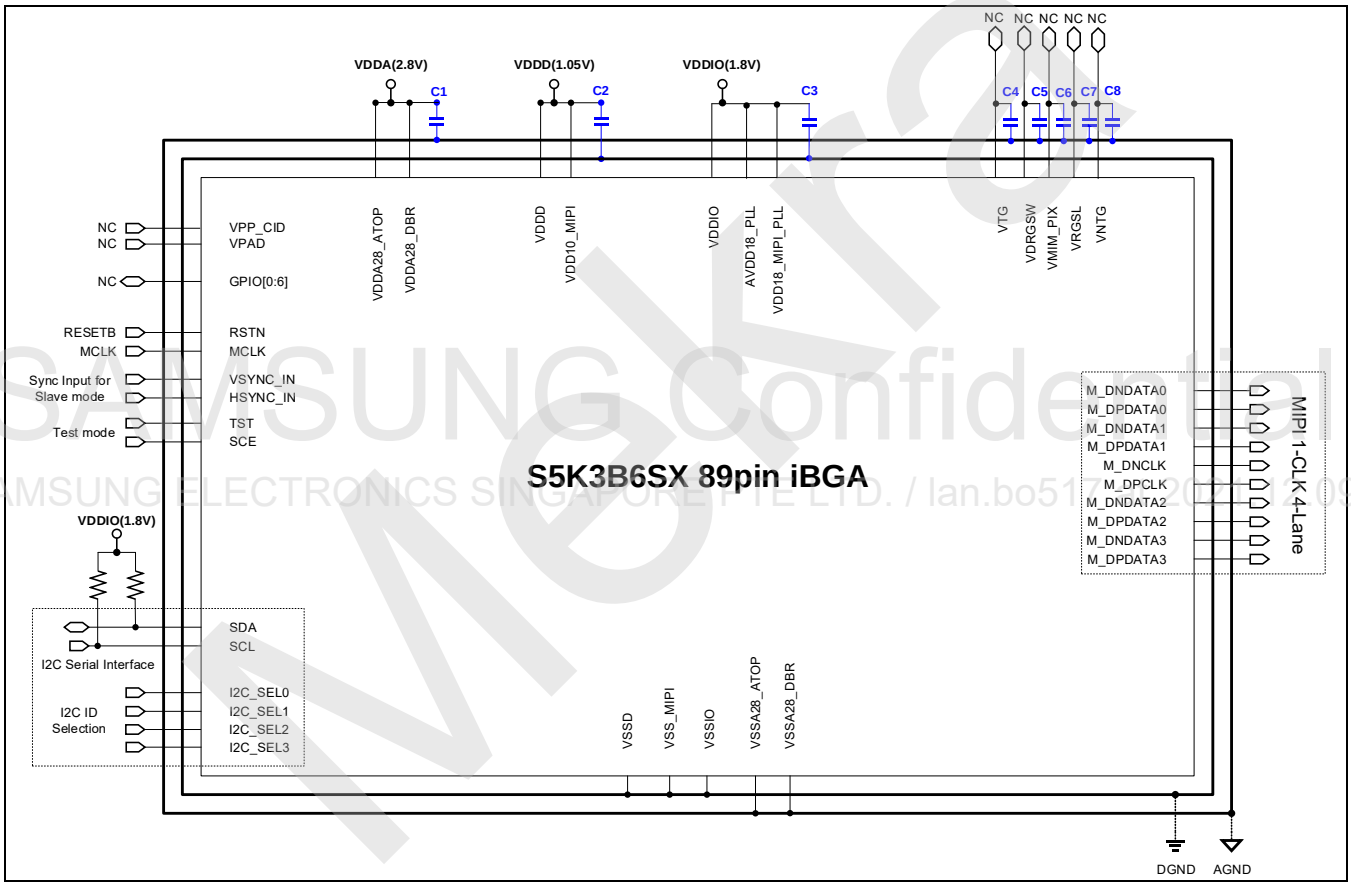


Figure 4 iBGA Application Circuit of MIPI

Table 2 Recommend Capacitor to Design a PCB(TBD)

No.	Power	Capacitor (μF)
C1	VDDA28_*	4.7 μF
C2	VDDD, VDD10_MIPI	2.2 μF
C3	VDDIO, AVDD18_PLL, VDD18_MIPI_PLL	2.2 μF
C4	VTG	0.1 μF

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C5	VDRGSW	0.1 µF
C6	VMIM_PIX	2.2 µF
C7	VRGSL	4.7 µF
C8	VNTG	4.7 µF

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3 Power sequence

This chapter includes the following sections:

- System state description and diagram
- Power-up/down sequence
- Standby/wake-up sequence
- Sensor system reset

3.1 System State

The Sensor module has four system state. Moving from one state to another is achieved by issuing the appropriate state command via the CCI serial control interface, the RSTN (XSHUTDOWN) signal changing state and the power supplies. By default, S5K3B6SX powers up with the CSI-2 serial data interface enabled. *Figure 5* defines the valid state changes for the sensor module.

S5K3B6SX support "Safety State" which ensure prohibit data stream when there might be any case of error occurred from error detection logic. Sensor move it's state from Safety state to Software standby if error cleared through registers. And after then, Sensor able to stream data.

If an error occurs during the data stream, S5K3B6SX outputs the last data normally and then moves to a Safety state.

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Table 3 System state summary

System State	Description
Power-off	Power supplies are turned off.
Hardware standby	RSTN is low. Power supplies are turned on, but could not communication with the sensor. Internal registers are set to default values.
Software standby	RSTN is high. Register setting through I2C is possible.
Streaming	The sensor is fully functional and video outputs are available.
Safety state	Sensor enters 'Safety state' when there could be fault in sensor operation. State which not able to stream data until resetting 'Error' flag. Goes to 'Software standby' statue after resetting 'Error' flag.

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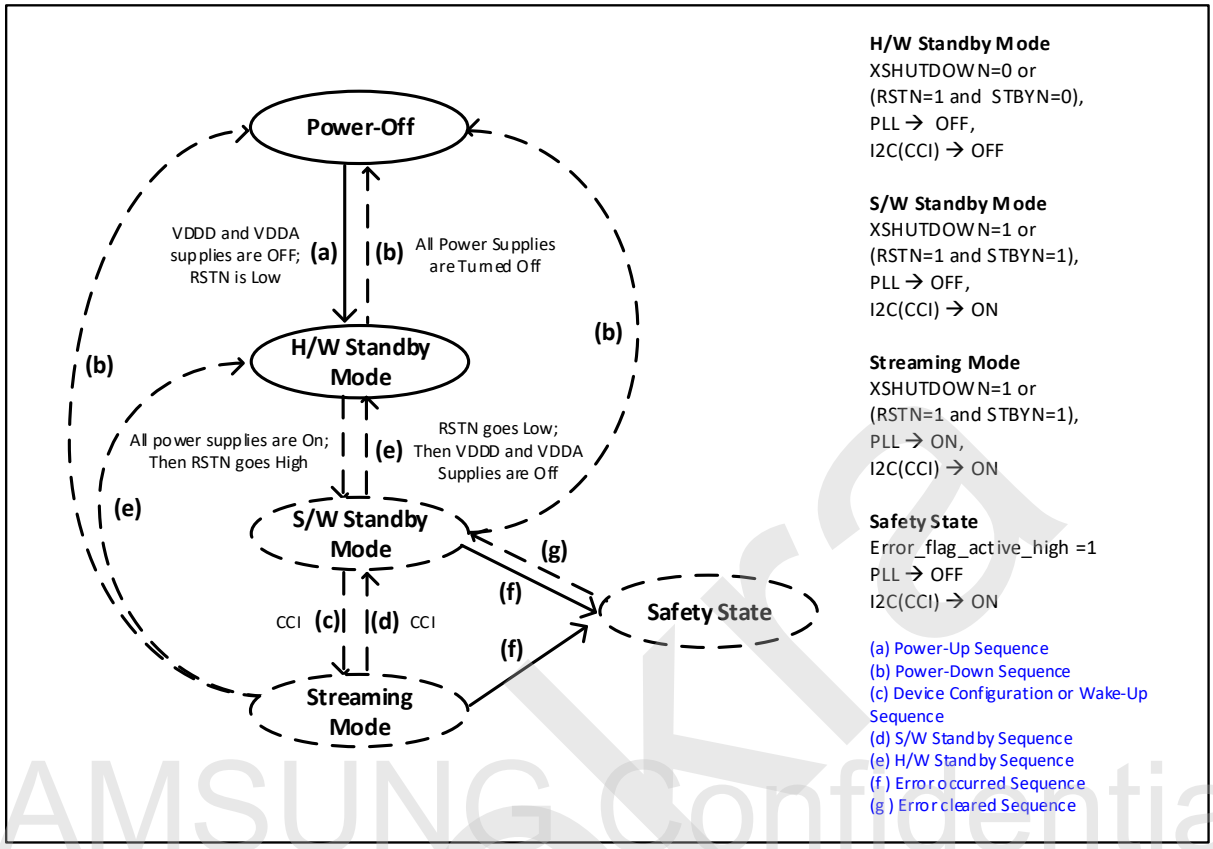


Figure 5 State Diagram of System

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3.2 Power-Up/Down Sequence

The digital and analog supply voltages can be powered up in any order, e.g., VDDD then VDDA or VDDA then VDDD.

On power up, RSTN (XSHUTDOWN) should be low when the power supplies are brought up, then the sensor module will go into hardware standby mode. As long as RSTN is low and VDDD is down, the sensor module stays in hardware standby mode.

The assertion of RSTN ensures that the CCI register values are initialized correctly to their default values.

When RSTN will go 'high', all PADs will exit from FAIL-SAFE mode, and switch to Normal operating mode.

The MCLK clock can either be initially low and then enabled during software standby mode or MCLK can be a free running clock.

NOTE:

1. At hardware standby mode, external VDDD and VDDA should be OFF.
2. External clock (MCLK) is recommended to be off in Software Standby to reduce power consumption.

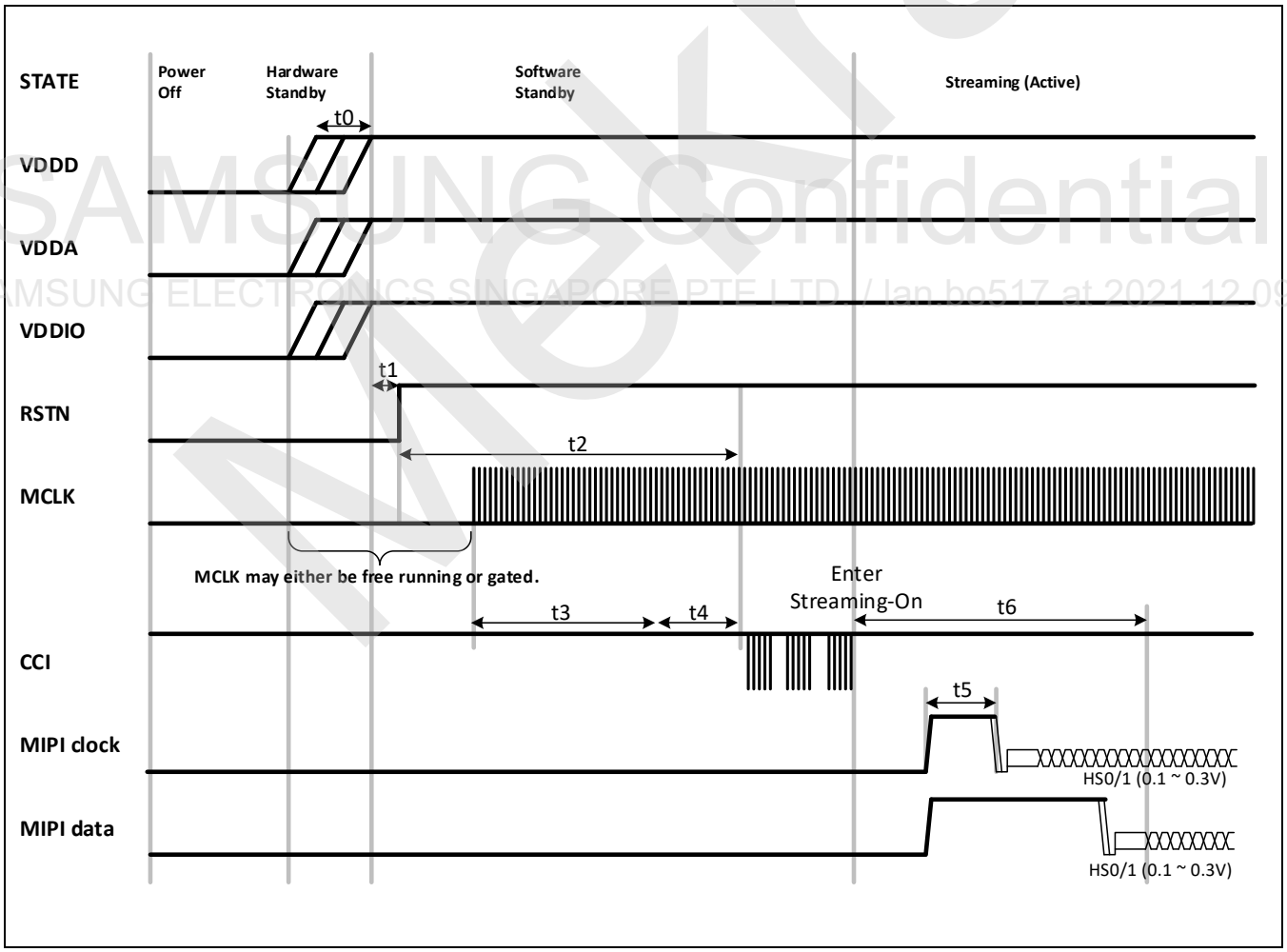


Figure 6 Power-Up Sequence Without Security IP

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Table 4 Power-up Sequence Timing Constraints

Constant	Label	Min.	Max.	Unit
VDDA//VDDD/VDDIO rising time	t0	VDDA/VDDD/VDDIO may rise in any order. The rising separation can vary from 0ns to indefinite.		ns
VDDD rising to RSTN (XSHUTDOWN) rising	t1	0.0	–	ns
RSTN (XSHUTDOWN) rising to first CCI transaction	t2	15.3	–	ms
RSTN rising to POST operation. * *POST - Power On Self Test for Memory (External Clock 27Mhz)	t3	10.83		ms
POST done to OTP operation	t4	4.5		Ms
DPHY initialization period (TINIT)	t5	0.1	–	ms
Entering streaming mode-The first frame start sequence	t6	+ the delay according to the coarse integration time value		–

TBD

Figure 7 Power-Up Sequence with Security IP

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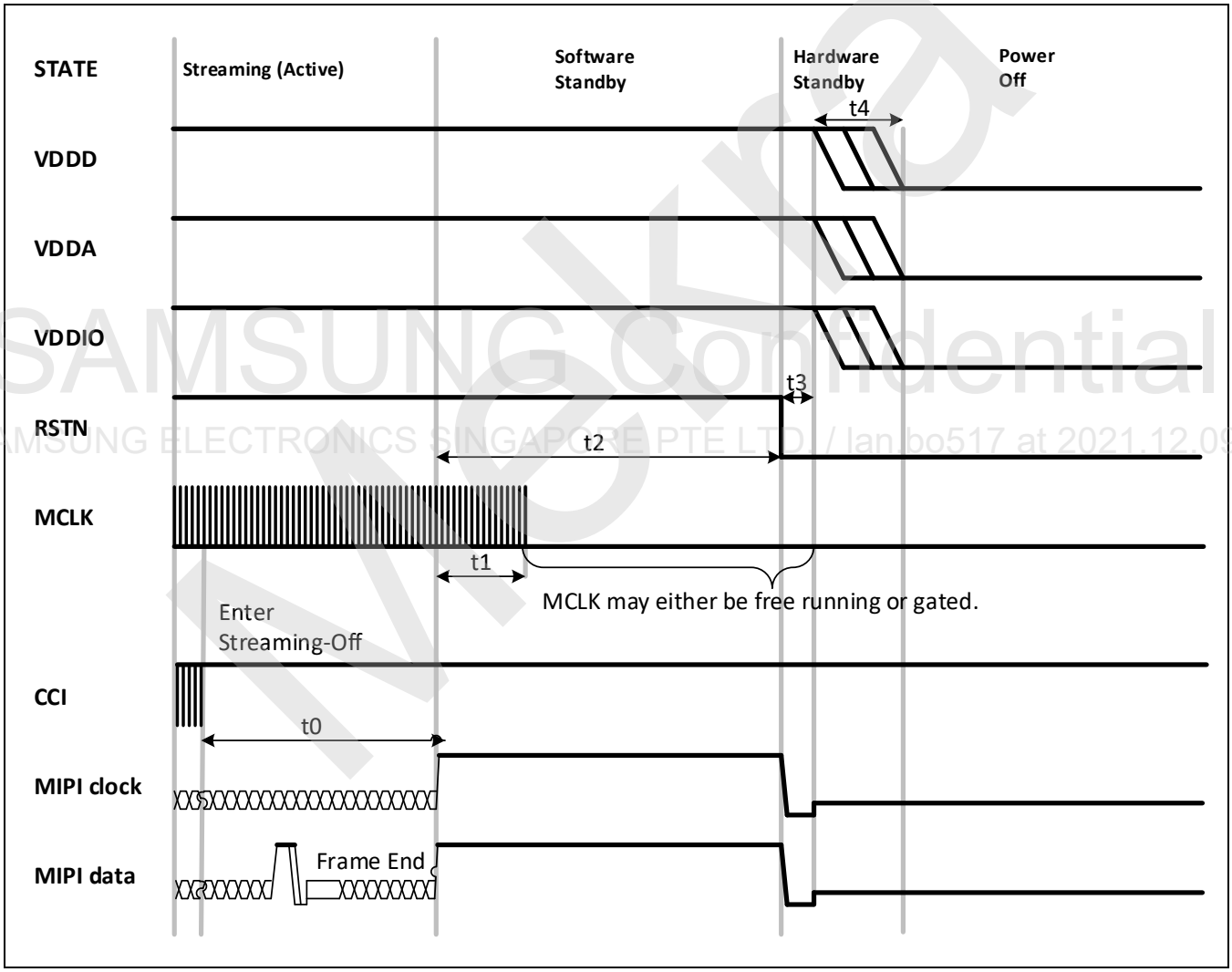
3.3 Power-Down Sequence

The digital and analog supply voltages can be powered down in any order, e.g., VDDD then VDDA or VDDA then VDDD.

Similar to the power-up sequence, the MCLK (EXTCLK): input clock may be either gated or continuous.

If the CCI command to exit streaming is received while a frame of valid active data is being output, then the sensor module must wait for the frame end code before entering software standby mode. Frame end code may come either after all frame pixels were transmitted or during the frame when next line transmission is completed - based on a configuration register (Refer to S5K3B6SX application notes for details)

If the CCI command to exit streaming mode is received during the inter frame time, then the sensor module must enter software standby mode immediately.



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Figure 8 Power-Down Sequence

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Table 5 Power-Down Sequence Timing Constraints

Constant	Label	Min.	Max.	Unit
Enter Software Standby CCI command- Device in Software Standby mode	t0	If outputting a frame of MIPI, the data waits for the MIPI frame end code before entering software standby; otherwise enter software standby mode immediately.		–
Minimum number of MCLK (EXTCLK) cycles after the last CCI transaction or MIPI frame end code.	t1	512	–	MCLK cycles
Last CCI Transaction or MIPI frame end code - RSTN (XSHUTDOWN) falling	t2	512	–	
RSTN (XSHUTDOWN) falling to VDDD or VDDA falling	t3	0.0	–	ns
VDDA/VDDA/VDDD/VDDIO falling time	t4	VDDA/VDDD/VDDIO may fall in any order. The rising separation can vary from 0 ns to indefinite		

NOTE: For minimal power at hardware standby mode external VDDD and VDDA should be OFF.

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4 Control Interface

This chapter includes the following sections:

- I2C
- CCI

4.1 I2C

S5K3B6SX supports the I2C communication method, which is the 2-wired serial interface between a backend processor (host) and a sensor, to control the register values.

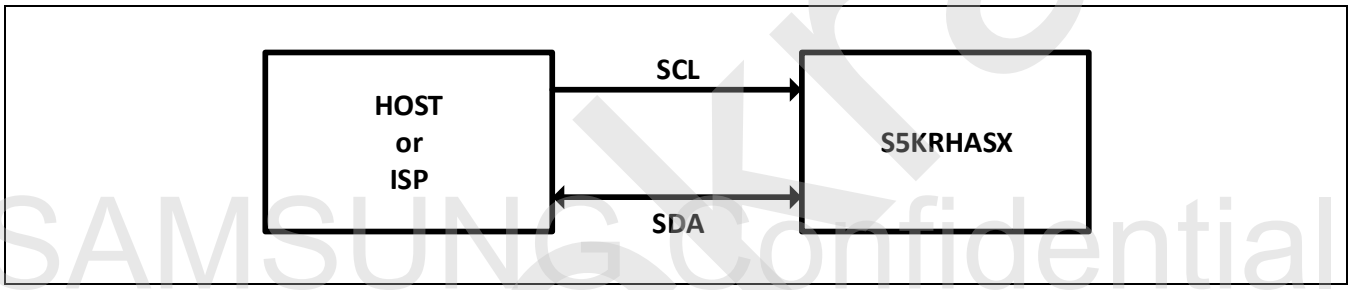


Figure 9 I2C Control Interface

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Table 6 describes the I2C signals.

Table 6 Pin Description of I2C

Pin Name	Description
SCL	Serial communication clock
SDA	Serial communication data

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4.2 CCI

S5K3B6SX supports the Camera Control Interface (CCI), which is an I2C Fast Mode+ compatible interface, to control the transmitter. S5K3B6SX always acts as a slave in CCI bus. CCI is capable of handling several slaves in the bus, but multi-master mode is not supported. Typically, only the receiver and transmitter are connected to CCI bus. This ensures a proper S/W implementation.

Typically, CCI is separated from system I2C bus, but I2C-compatibility ensures that it can connect the transmitter to system I2C bus. CCI is a subset of the I2C protocol including the minimum combination of obligatory features for the I2C slave device specified in the I2C specification. Therefore, transmitters complying with CCI specification are connected to system I2C bus. However, it is important to ensure that the I2C masters do not try to utilize these I2C features, which are not supported in transmitters that are complying with CCI specification. Each transmitter conformed to CCI specification has additional features to support I2C.

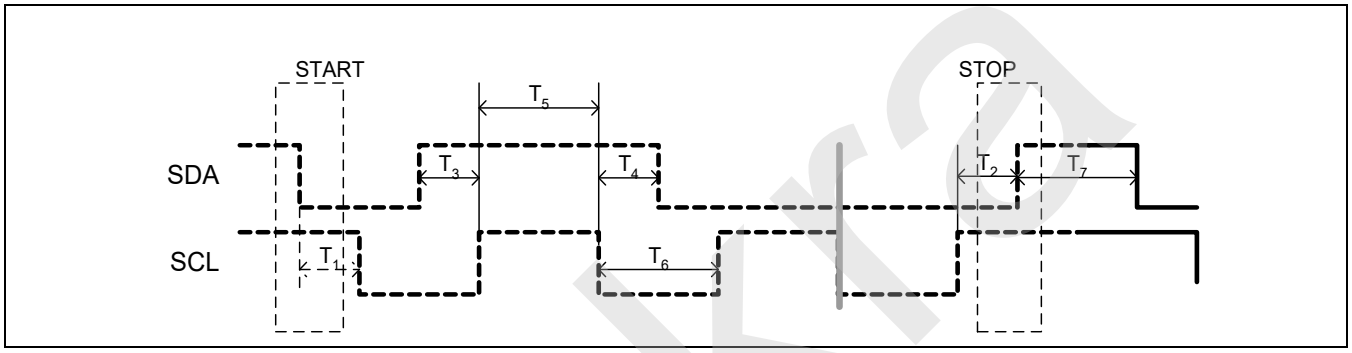


Figure 10 Timing Diagram of CCI

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Table 7 I2C Standard Mode Timing Specifications

Parameter	Symbol	Min.	Max.	Unit
SCL clock frequency	-	0	100	kHz
Hold time for start condition	T_1	4.0	-	µs
Setup time for stop condition	T_2	4.0	-	
Data setup time	T_3	250	-	ns
Data hold time	T_4	0	3.45	µs
High period of the SCL clock	T_5	4.0	-	
Low period of the SCL clock	T_6	4.7	-	
Bus free time between stop and start conditions	T_7	4.7	-	
Rise time for both SDA and SCL signals	-	-	1000	ns
Fall time for both SDA and SCL signals	-	-	300	
Capacitive load for each bus line	CB	-	400	pF

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Table 8 I2C Fast Mode Timing Specifications

Parameter	Symbol	Min.	Max.	Unit
SCL clock frequency	-	0	400	kHz
Hold time for start condition	T_1	0.6	-	µs
Setup time for stop condition	T_2	0.6	-	
Data setup time with external clock (MCLK) above 12.8 MHz	T_3	0.1	-	µs
Data setup time with external clock (MCLK) below 12.8 MHz		0.6	-	
Data hold time	T_4	0	0.9	µs
High period of the SCL clock	T_5	0.6	-	
Low period of the SCL clock	T_6	1.3	-	
Bus free time between stop and start conditions	T_7	1.3	-	
Rise time for both SDA and SCL signals	-	-	300	ns
Fall time for both SDA and SCL signals	-	-	300	
Capacitive load for each bus line	CB	-	400	pF

Table 9 I2C Fast Mode+ Timing Specifications

Parameter	Symbol	Min.	Max.	Unit
SCL clock frequency	-	0	1000	kHz
Hold time for start condition	T_1	0.26	-	µs
Setup time for stop condition	T_2	0.26	-	
Data setup time, external clock (MCLK) above 12.8 MHz	T_3	0.6	-	µs
Data hold time	T_4	0	-	µs
High period of the SCL clock	T_5	0.26	-	
Low period of the SCL clock	T_6	0.5	-	
Bus free time between stop and start conditions	T_7	0.5	-	
Rise time for both SDA and SCL signals	-	-	120	ns
Fall time for both SDA and SCL signals	-	-	120	
Capacitive load for each bus line	CB	-	550	pF

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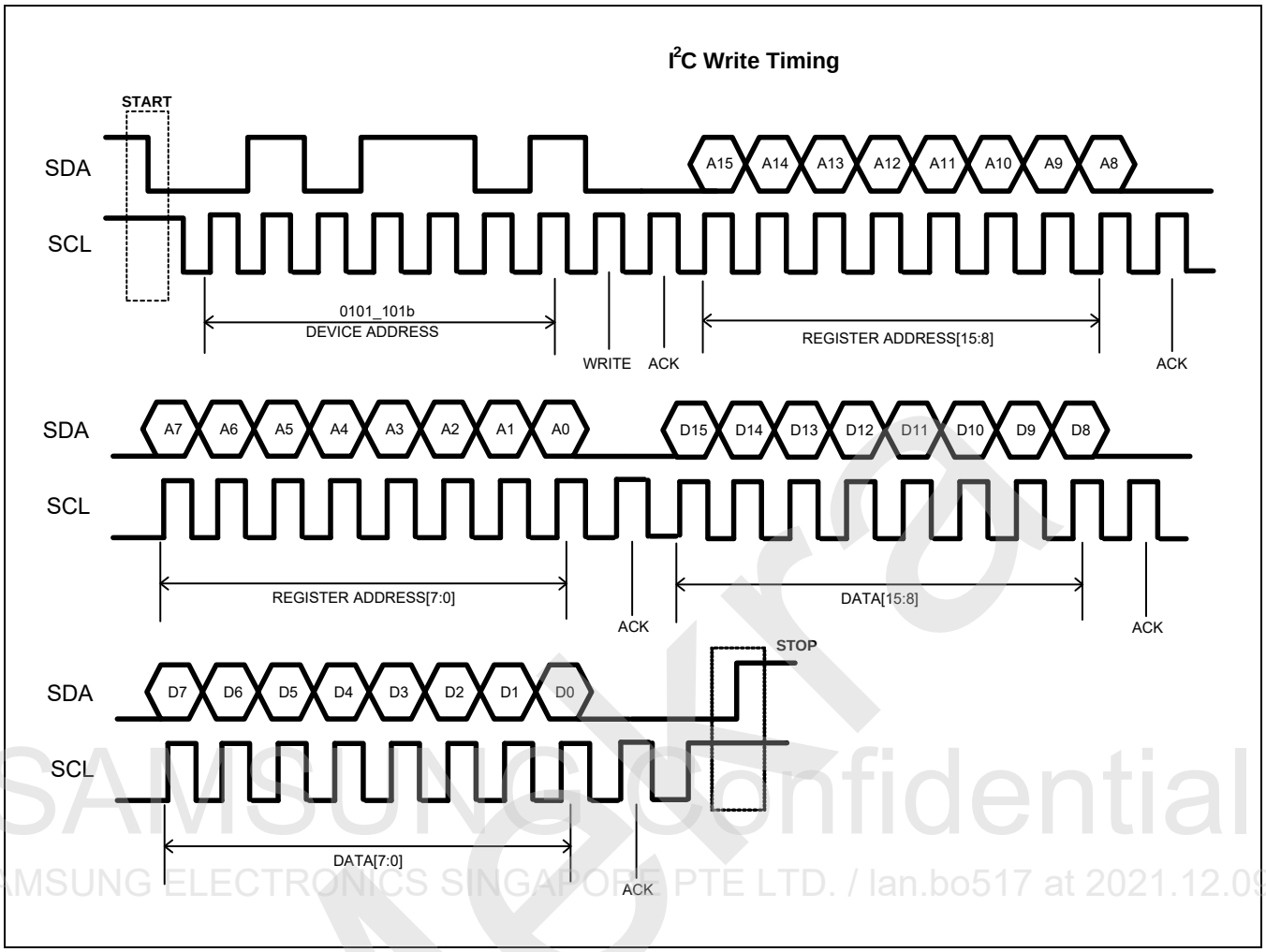


Figure 11 CCI Write Timing Example

NOTE: The device address can be changed by pin configuration of I2C_SEL, as described in the pad description.

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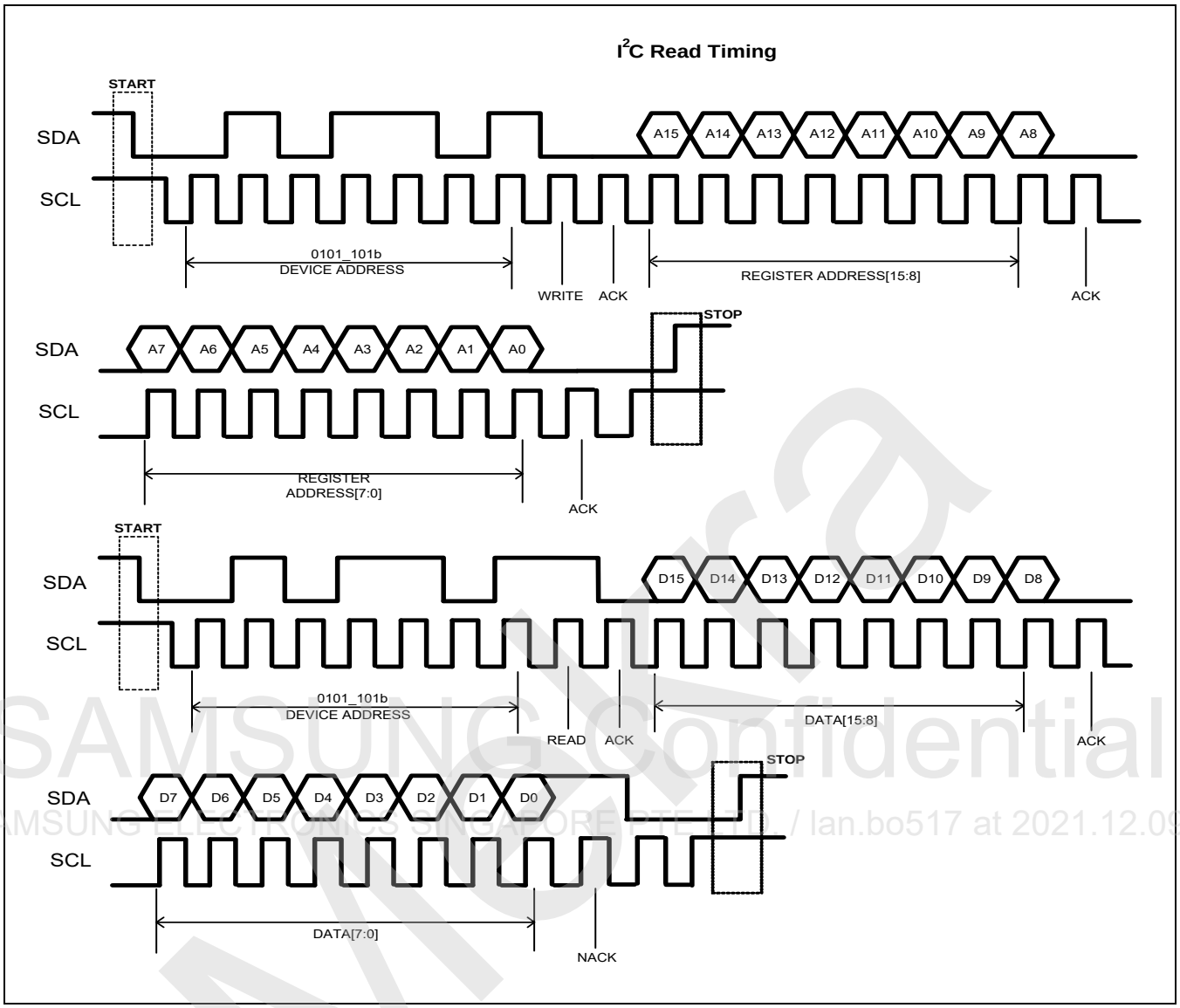


Figure 12 CCI Read Timing Example

NOTE: The device address can be changed by the pin configuration of I2C_SEL, as described in the pad description.

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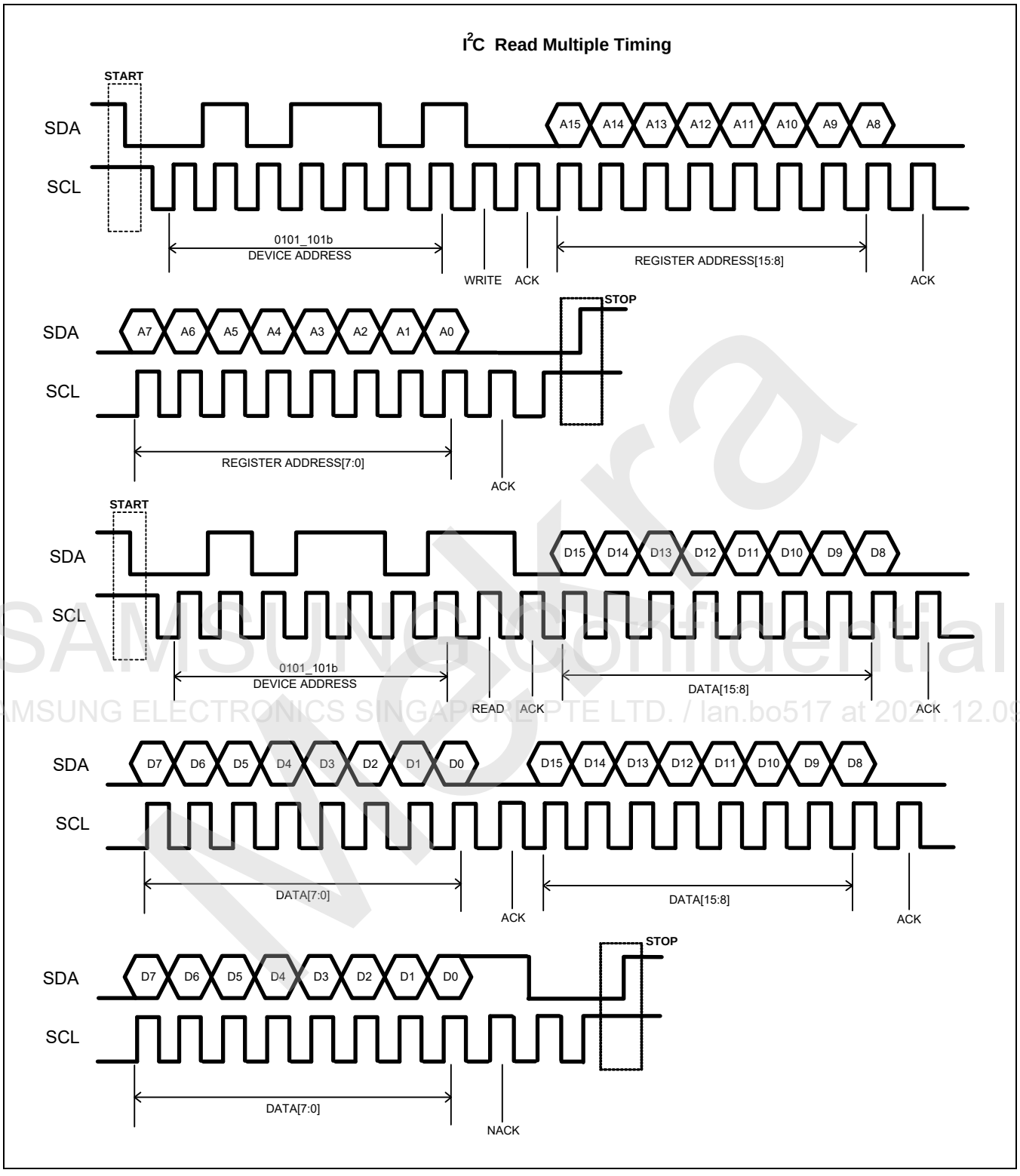


Figure 13 CCI Read Multiple Timing Example

NOTE: The device address can be changed by the pin configuration of I2C_SEL, as described in the pad description.

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It is possible to configure up to eight I2C slave addresses using I2C_SEL3 ~ I2C_SEL0 pins.

Table 10 I2C ID Address (I2C_SEL3 ~ I2C_SEL0 Pad)

I2C_SEL[3:0]	Slave Address (7-bit + Read Mode)	Slave Address (7-bit + Write Mode)	Comment
0	0010_0001b/21h	0010_0000b/20h	Address 1
1	0011_0001b/31h	0011_0000b/30h	Address 2
2	0010_1001b/29h	0010_1000b/28h	Address 3
3	0011_1001b/39h	0011_1000b/38h	Address 4
4	0110_0001b/61h	0110_0000b/60h	Address 5
5	1010_0001b/A1h	1010_0000b/A0h	Address 6
6	1100_1001b/C9h	1100_1000b/C8h	Address 7
7	1110_1011b/EBh	1110_1010b/EAh	Address 8

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5 Pixel Information

This chapter includes the following sections:

- Pixel array
- Spectral sensitivity characteristics

5.1 Pixel Array

The effective pixel array is 1920 (H) × 1536 (V). Although the total pixel array includes different kinds of pixels, pixel array can be read from (8, 48) to (1927, 1583). The pixels are covered by clear and red color filters arranged in RGB Bayer structure. Micro-lens are placed on the top of color filters to increase the effective fill factor for higher sensitivity.

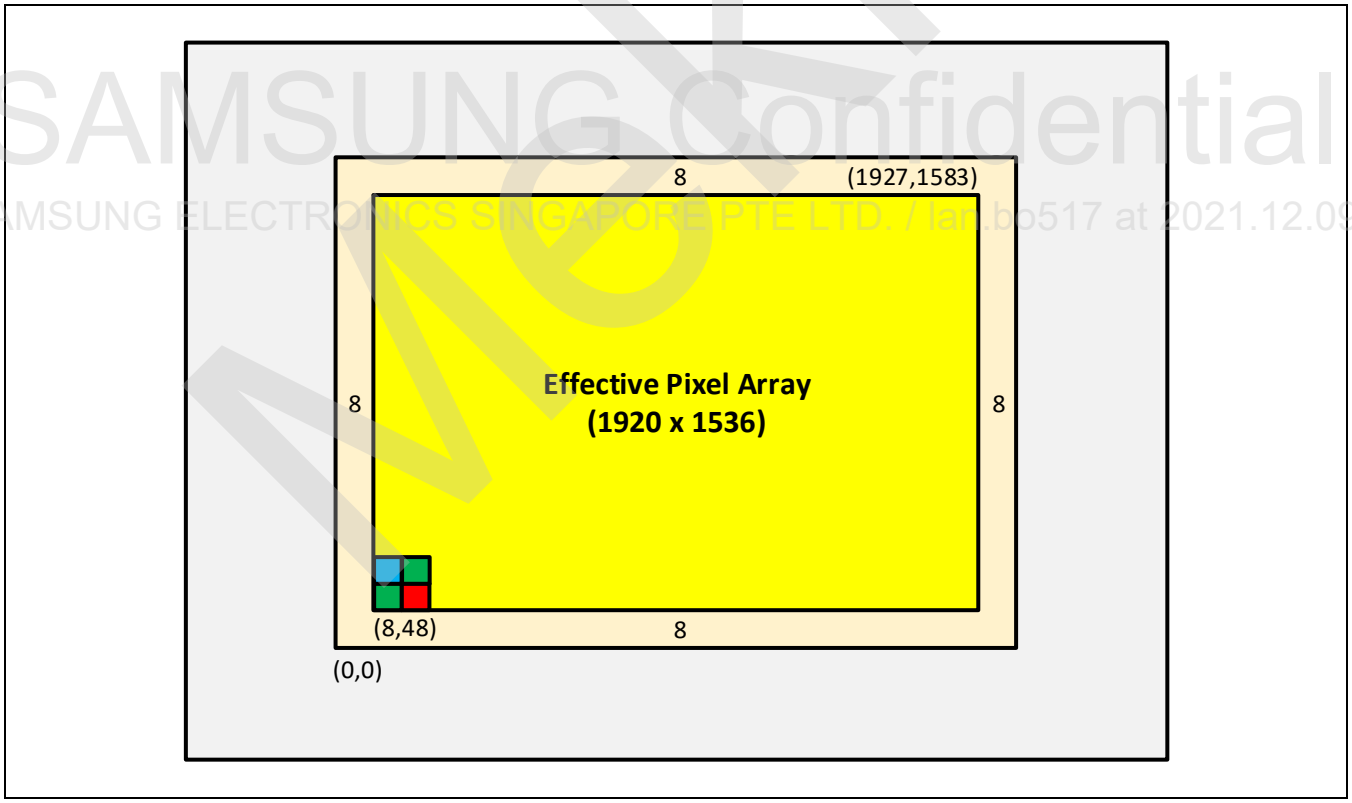


Figure 14 Pixel Array Information

NOTE: For top view of the chip, vertically flip the displayed image.

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5.2 Spectral Sensitivity Characteristics

The spectral sensitivity of S5K3B6SX is characterized at the half code of clear signal on wavelength sweep.



Figure 15 Spectral Sensitivity Characteristics

NOTE: For any improvement, it can change without any notice before the mass-production.

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6 Functional Description

This chapter includes the following sections:

- Functional Block Diagraim
- PLL and clock generator
- Operation mode
- Slave mode
- Frame readout format
- Image orientation (H/V)
- Frame rate control
- Integration time control (Electronic shutter control)
- Gain control
- Auto dark level correction
- Image output format

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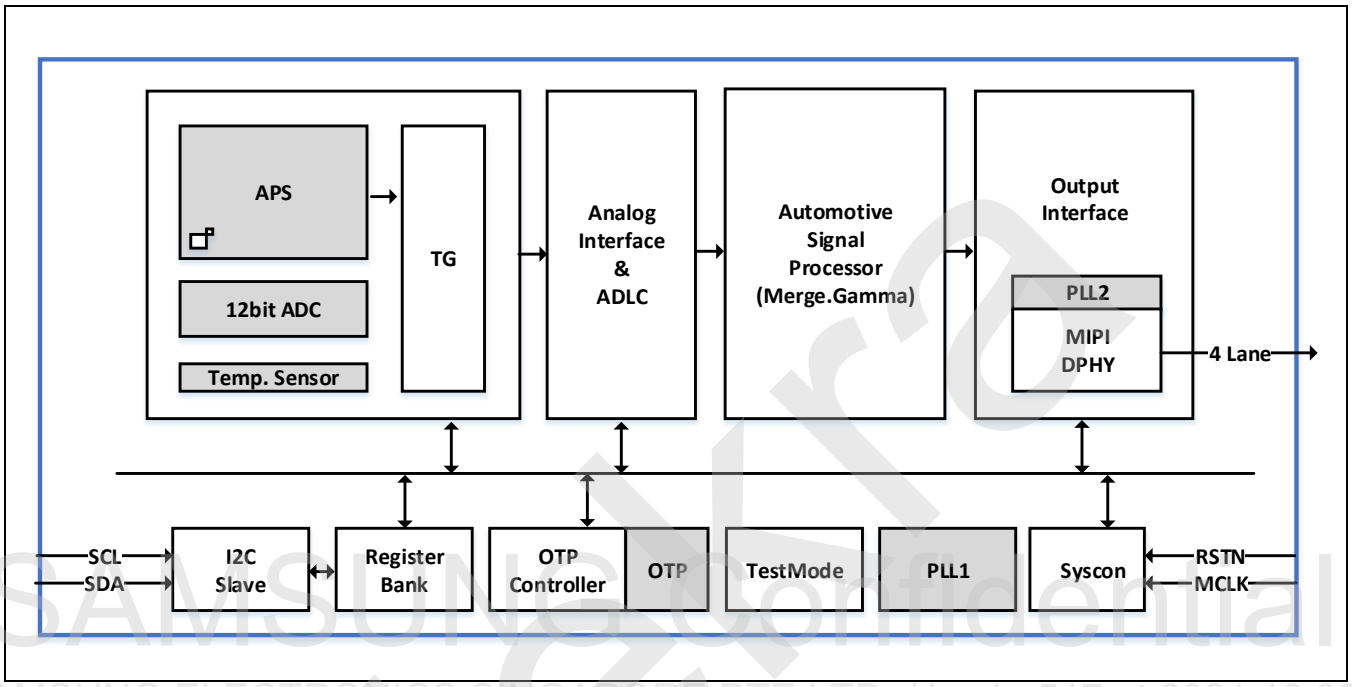
S5K3B6SX provides several readout modes and specific functions for video systems and for the required variety of clocks. Two Phase-Locked Loops (PLLs) are embedded in the sensor. One generates clocks for systems, and the other one generates clocks for high-speed serial interfaces. The master input clock range is between 6 MHz and 64 MHz.

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6.1 Functional Block Diagram

S5K3B6SX is composed of an APS array, 12-bit ADCs, an embedded Timing Generator (TG), pixel data process blocks, OTP, two Phase-Locked Loop of PLL1 and PLL2, Automotive Signal Processing block which support interleaving, Merge and Compress for WDR data and interface blocks.

Figure 16 illustrates the functional block diagram of sensor.



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Figure 16 Functional Block Diagram of Sensor

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6.2 PLL and Clock Generator

S5K3B6SX contains two PLLs and a clock generator, which generate all the necessary video timing, output pixel clocks, and interface serializing clocks.

By setting the divide-ratio for pre PLL clock divider (`pll_sys_p and pll_int_p`), PLL multiplier (`pll_sys_m and pll_int_m`), and post scaler (`pll_sys_s and pll_int_s`) appropriately, you can get the necessary PLL output clocks (`PLL_OUT1 and PLL_OUT2`).

PLL can handle any external input clock in the range of 6 MHz to 64 MHz (27 MHz is default), and synthesize PLL_OUT1 between 40 MHz to 2500 MHz and PLL_OUT2 between 70.3125 MHz to 4500 MHz. For proper PLL operation, PLL input clock (`PLL_IN1 and PLL_IN2`) should be in the range of 6 MHz to 12 MHz. All PLL programming should be performed during software standby mode for stable system operation.

MIPI output is DDR using MIPI word clock, so that MIPI word clock is 1/16th of the bit clock. *Figure 17* illustrates the overall clock tree structure.

Interface PLL(PLL2) of S5K3B6SX is able to support SSCG(Spread Spectrum Clock Generation) to avoid EMI issues.

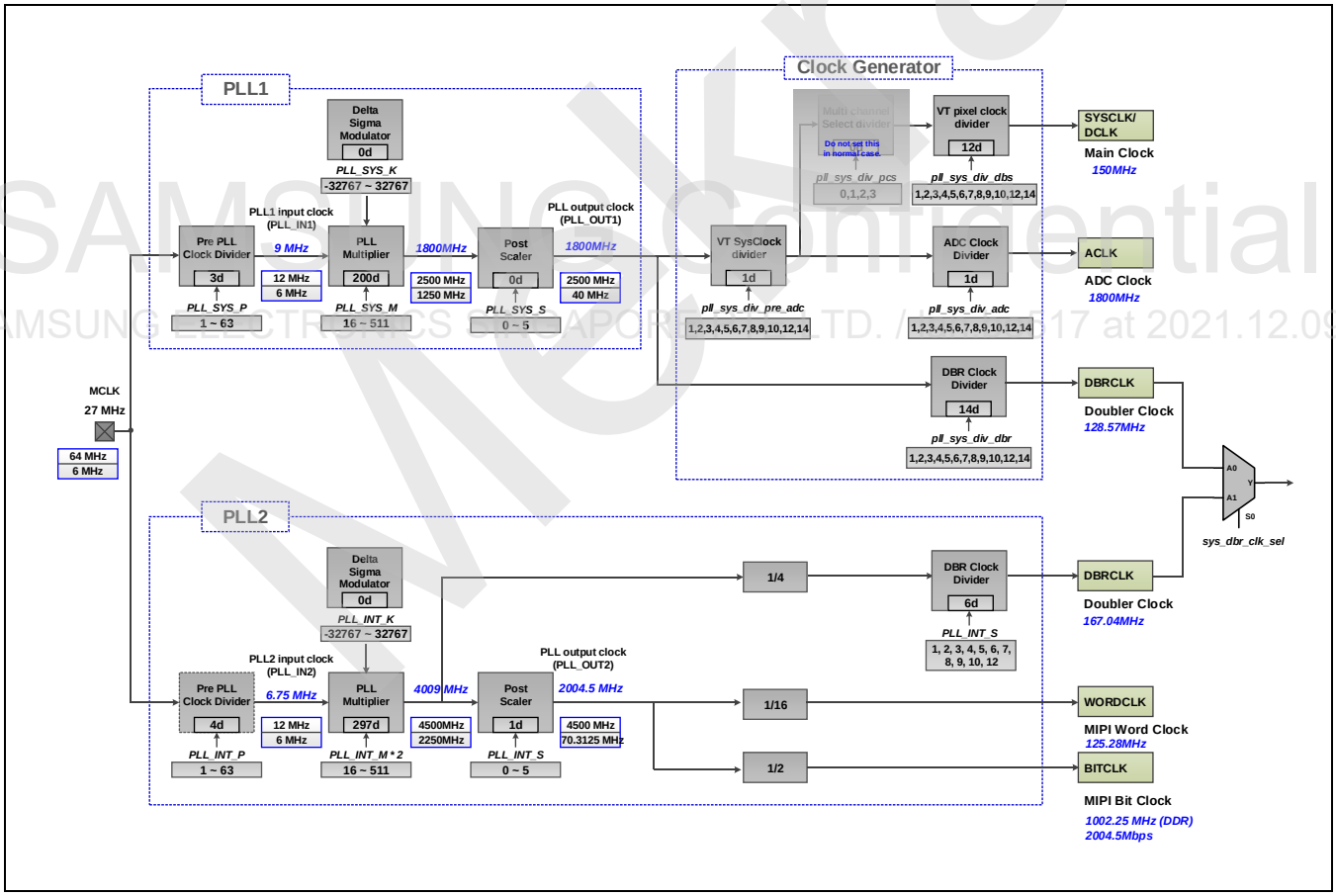


Figure 17 PLL and Clock Generator

NOTE:

All the register setting values are represented in decimal unit.

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Use the following equations to synthesize all necessary frequencies.

PLL1 Input Clock = External Input Clock (MCLK)/pll_sys_p (6 MHz to 12 MHz)
PLL2 Input Clock = External Input Clock (MCLK)/pll_int_p (6 MHz to 12 MHz)
PLL_OUT1 = (PLL_IN1×(pll_sys_m+pll_sys_k/65536))/2^pll_sys_s (40 MHz to 2500 MHz)
PLL_OUT2 = (PLL_IN2×(pll_int_m+pll_int_k/65536)*2)/2^pll_int_s
(70.3125MHz to 400 MHz, Bit clock divided by 2 for DDR)
1 ≤ pll_*_p ≤ 63 : 16 ≤ pll_*_m ≤ 511 : 0 ≤ pll_*_s ≤ 5 : -32767 ≤ pll_*_k ≤ 32767

6.3 Operation Mode

S5K3B6SX supports the operation modes described in *Table 11* along with their speed and output size.

Table 11 Typical Operation Modes

Mode	Interface	Output format	ADC [Bit]	Frame Rate [fps]	Bit Rate (Per Lane)	H-Pixel [ea]	V-Pixel [ea]	Comment
20b Combined	MIPI 4Lane	RAW20	12	60	1280	1936	1552	Intra-scene DCG*2) with LFM
16b Combined and Compress		RAW16	12	60	960	1936	1552	
14 Combined and Compress		RAW14	12	60	960	1936	1552	
12 Combined and Compress		RAW12	12	60	800	1936	1552	
Line Re-order		RAW12	12	45	1760	1936	4656	
Pixel Re-order		RAW12	12	45	1760	5808	1552	
24b Combined		RAW12^2 *3)	12	60	1280	1936	1552	Intra-scene DCG*2) without LFM
20b Combined and Compress		RAW20	12	60	1120	1936	1552	
16b Combined and Compress		RAW16	12	60	960	1936	1552	
14 Combined and Compress		RAW14	12	60	800	1936	1552	
12 Combined and Compress		RAW12	12	60	640	1936	1552	
Line Re-order		RAW12	12	45	2080	1936	6208	
Pixel Re-order		RAW12	12	45	1920	7744	2186	

NOTE: 1. To improve the performance, the values of operation modes, output size, frame rate, and so on can be changed.

NOTE: 2. Sensor readout same pixel twice with difference conversion gain in frame.

NOTE: 3. The 24b Combined frame output in the RAW12 format where the horizontal pixel size doubled due to there is no defined output format of RAW24 in MIPI alliance.

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6.4 Slave Mode

To support multiple camera system, S5K3B6SX can operate in slave mode using the external sync. Vertical sync signal and the second CCI slave ID. Depending on the operation mode of the system, the direction of 'VSYNC_IN' PAD can be controlled as output or input mode.

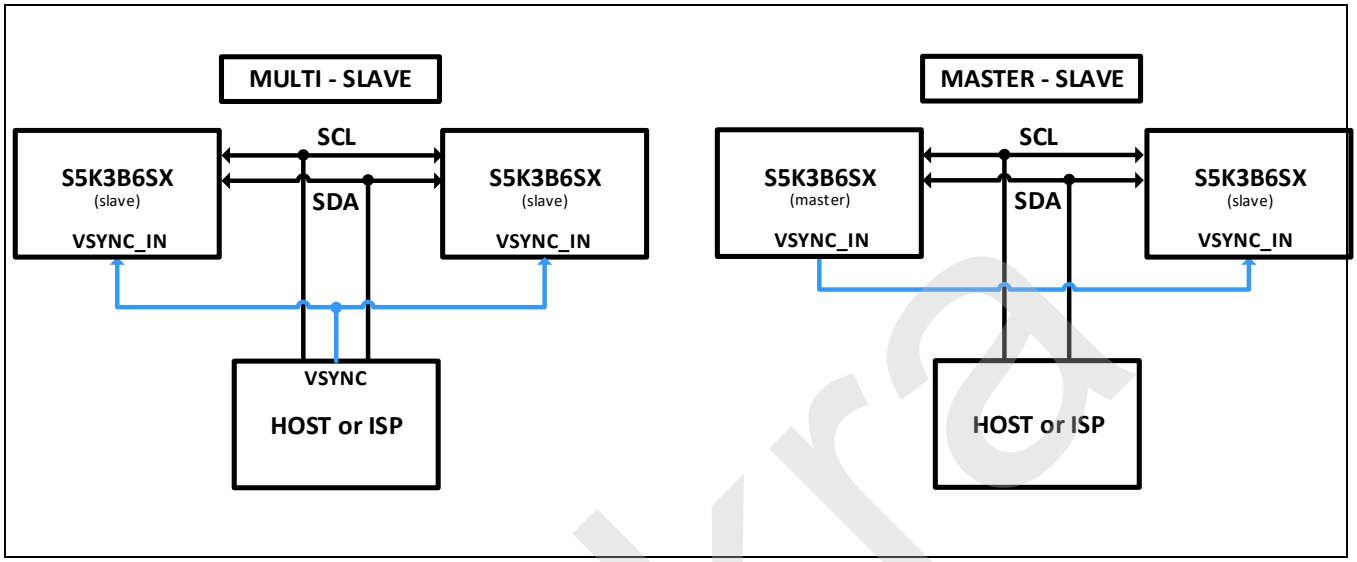


Figure 18 System Diagram for Multiple Camera

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6.5 Frame Readout Format

S5K3B6SX outputs up to 4 ea different (LPD HCG, LPD LCG, SPD, and LPD short) frames through the line-interleaved WDR operation. Intra-scene DCG without LFM mode supports the maximized dynamic range after the recombination in an internal function or external SoC. During the WDR readout operation, the 1-row readout time of each exposed frame is tuned, and then the data of LPD HCG, LPD LCG, SPD, and LPD short frame is out in every 4-row readout time period. This sensor can also operate in Intra-scene DCG with LFM mode.

6.5.1 Data Readout Frame Format

S5K3B6SX supports various data types and frame formats. *Figure 19* illustrates two kinds of output frame format. In line interleaving, LPD HCG, LPD LCG, SPD, and LPD short row line data output in sequential order at every one row data output time. But in pixel reorder, LPD HCG, LPD LCG, SPD, and LPD short data aligned and output in sequential pixel interleaving. In this case, whole pixel data in the same row is output over four times of the one row data output time in line interleaving.

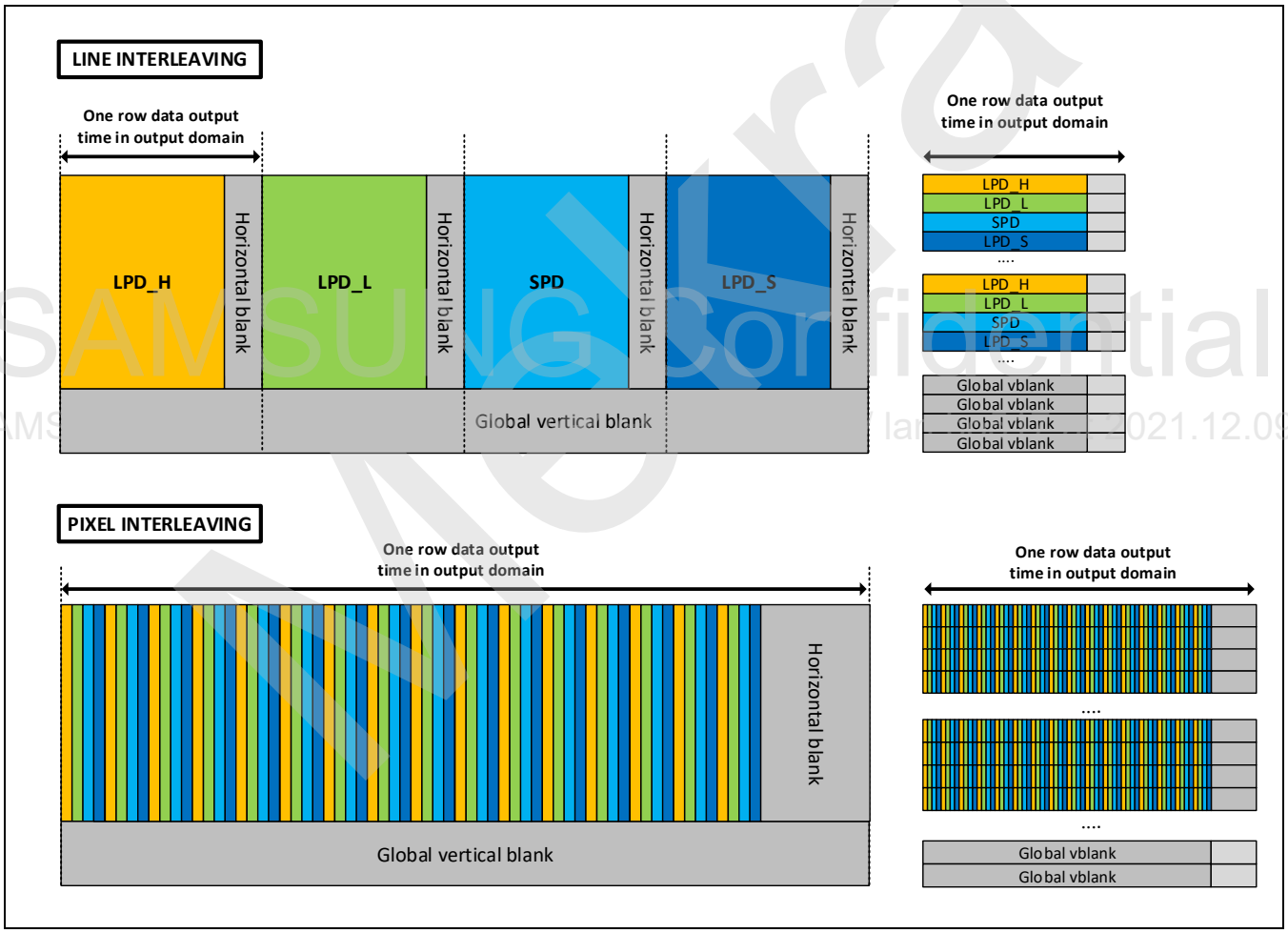


Figure 19 Line/Pixel Interleaved WDR Output Frame Format (Intra scene DCG without LFM)

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6.5.2 Data Combine

To reduce the amount of data transmission, S5K3B6SX can output combined 24-bit data instead of 4x12 bit raw data. Four 12-bit data are normalized to 24 bit by by using the internal algorithm.

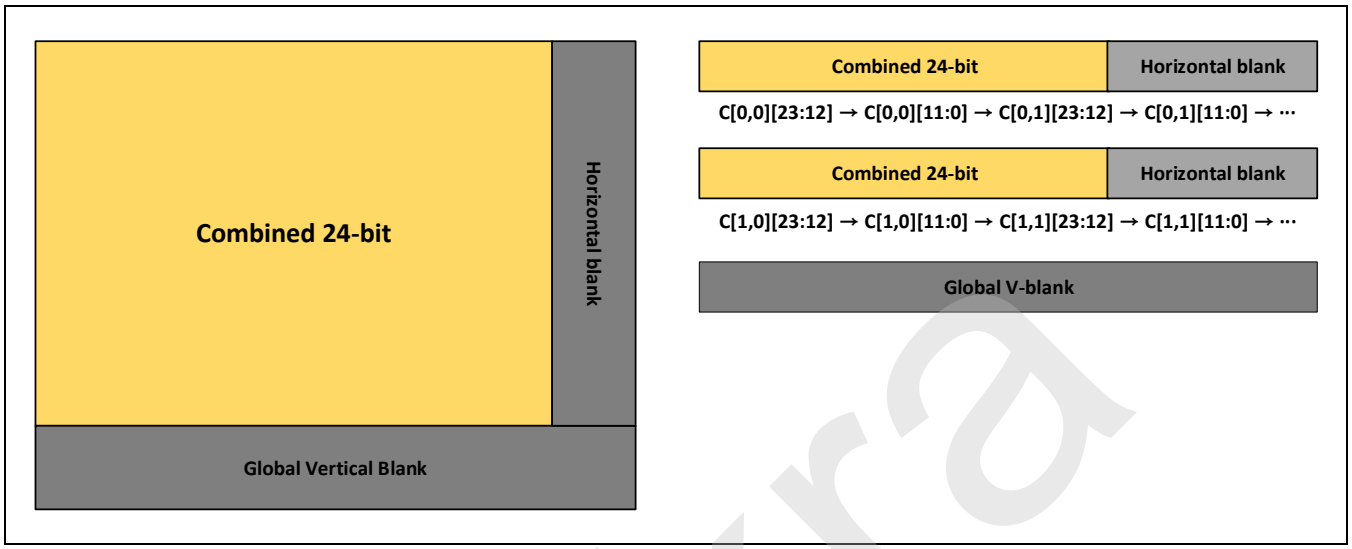
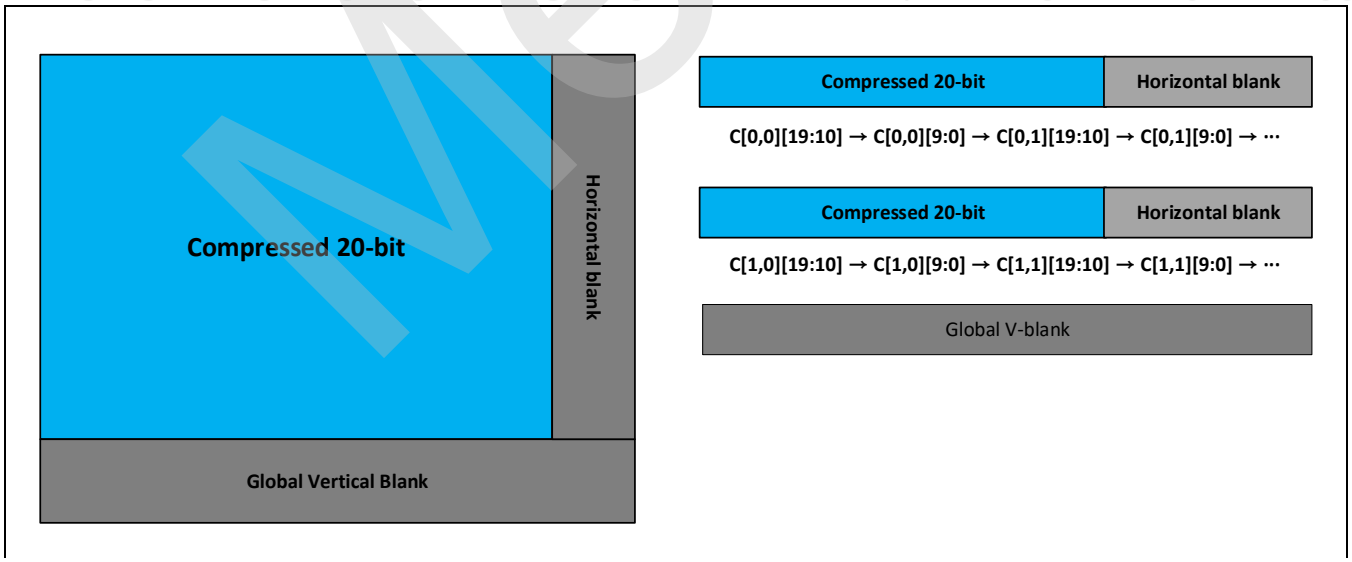


Figure 20 Combined Data Output Frame Format (Intra-scene DCG without LFM mode)

6.5.3 Data Compress

S5K3B6SX supports 20/16/14/12-bit compressed data output mode. The piecewise linear compression compresses the 24-bit combined data to 20/16/14/12-bit data with 32 knee-points. These knee-points are controlled by registers.

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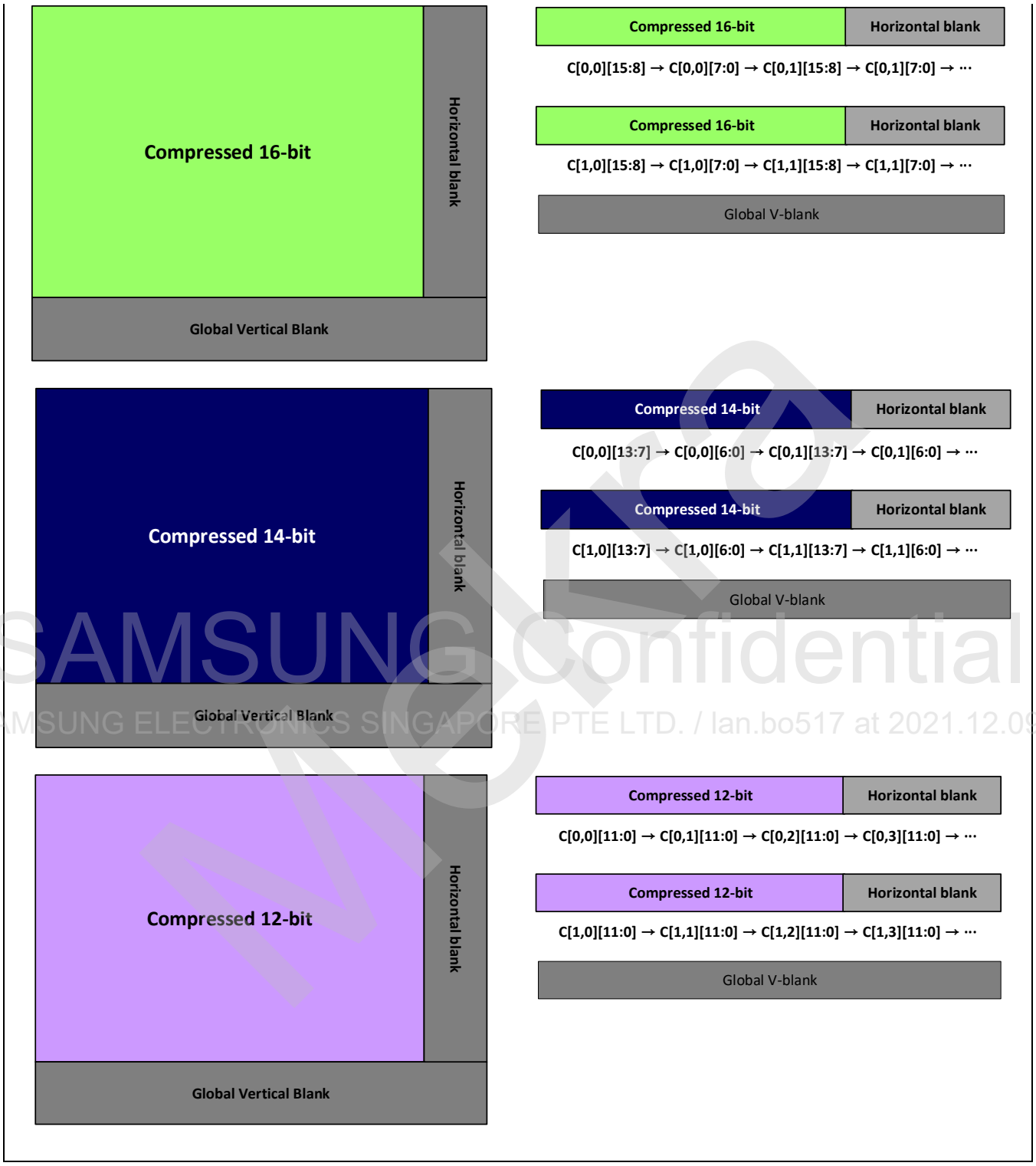


Figure 21 20/16/14/12-Bit Compressed Data Output Format (Intra-scene DCG without LFM mode)

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6.6 Image Orientation (H/V)

The pixel data are normally readout from left to right in horizontal direction and from top to bottom in vertical direction. By changing the mirror/flip mode, the readout order is reversed and the resulting image is flipped as a mirror image. Pixel data outs from right to left in horizontal mirror mode and from bottom to top in vertical flip mode. The horizontal mirror and the vertical flip modes are programmed by the image orientation register.

Following are the four pixel readout orders supported by sensor:

- **Standard readout:** Horizontal start address and width registers control the addressed region of horizontal pixel data, and vertical start address and width registers control the addressed region of vertical pixel data.
- **Horizontally mirrored readout:** Horizontal start address and orientation registers control the addressed region of horizontal pixel data output, and the addressed region of vertical pixel data output is the same as that of the standard readout.
- **Vertical flipped readout:** The addressed region of horizontal pixel data output is same as that of the standard readout, and vertical start address and orientation registers control the addressed region of the vertical pixel data output.
- **Horizontally mirrored and vertically flipped readout:** Horizontal start address and orientation registers control the addressed region of the horizontal pixel data, and vertical start address and orientation registers control the addressed region of the vertical pixel data.

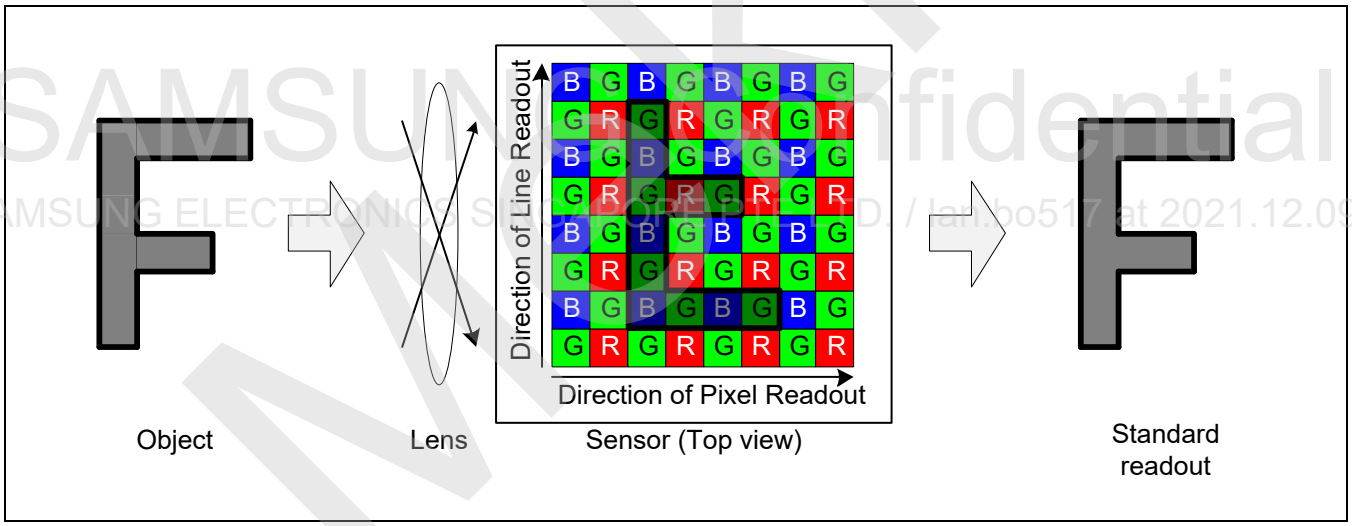


Figure 22 Object, Sensor and displayed image in Standard Readout

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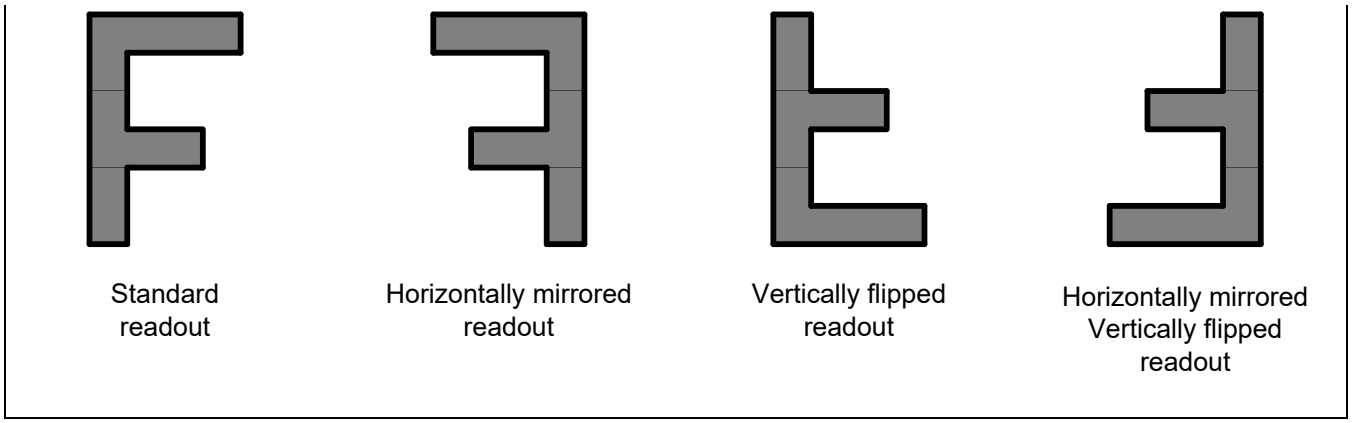


Figure 23 Four readout orders and displayed images

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6.7 Frame Rate Control

Change the size of the virtual frame to change the line rate and frame rate. The number of clocks for horizontal readout and number of lines for vertical readout that consist of the virtual frame are controlled by registers. The horizontal and vertical blanking times (horizontal blanking time: # of clocks for horizontal readout columns, vertical blanking time: # of lines for vertical readout lines) must meet the system requirements.

Use the following equation to calculate the frame rate.

Frame rate = SYSCLK / (# of clocks for horizontal readout[1)] * # of lines for vertical readout[2)])

[1)] # of clocks for horizontal readout = line_length_pck
[2)] # of lines for vertical readout = frame_length_lines ("frame length lines" value should divide into 3 at intra-scene DCG with LFM mode and 4 in the intra-scene DCG without LFM mode.)

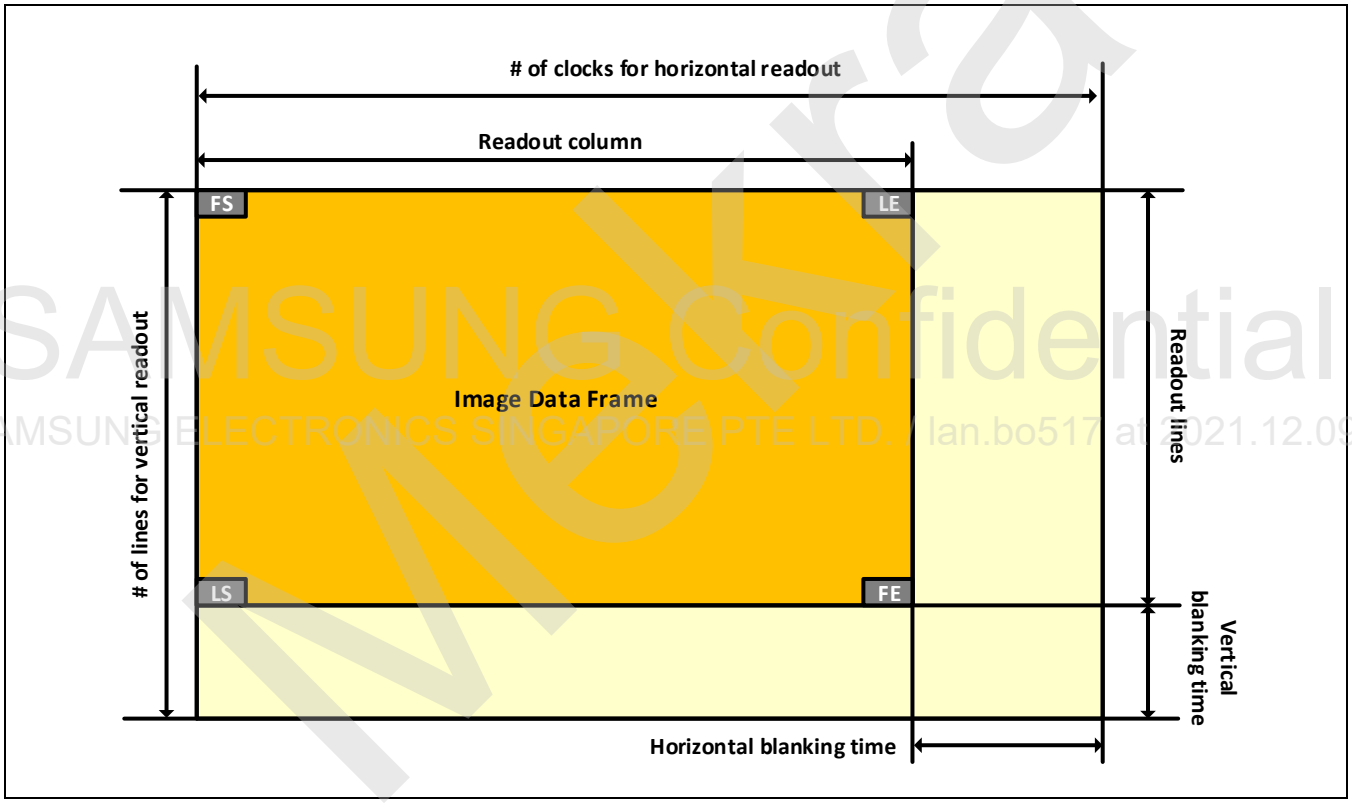


Figure 24 Example Format of Virtual Frame

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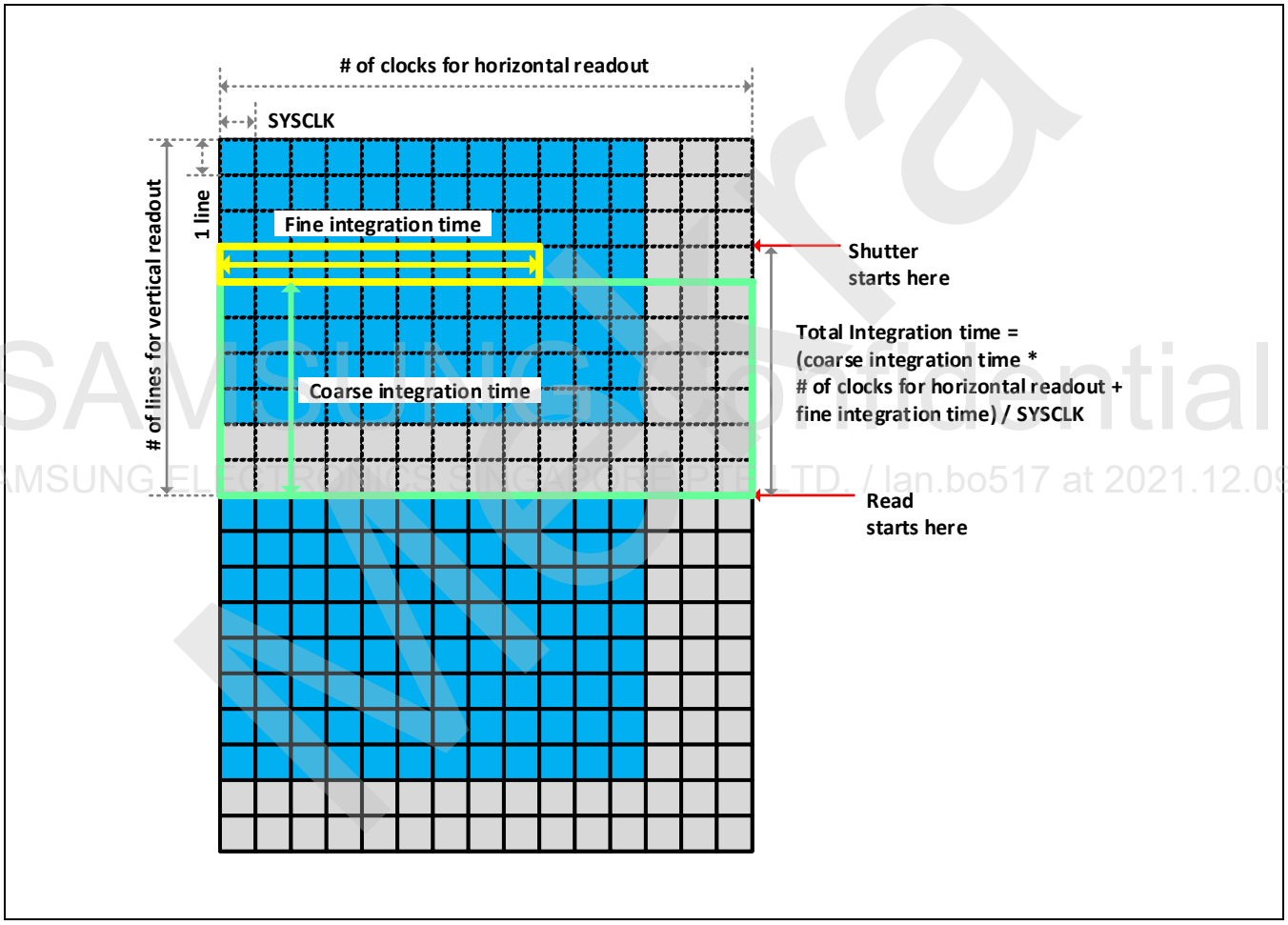
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6.8 Integration Time Control (Electronic Shutter Control)

The pixel integration time is controlled by the shutter operation. During the shutter operation, the amount of integration time is determined by the column step integration time control register (fine_integration_time) and the line step integration time control register (coarse_integration_time). Use the following formula to calculate the total integration time of the sensor.

```
Total_integration_time = {coarse_integration_time x # of clocks for horizontal readout[1)]+
fine_integration_time}/(SYSCLK) [sec]

[1)] # of clocks for horizontal readout = line_length_pck
```



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Figure 25 Example of Calculating Integration Time

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6.9 Gain Control

6.9.1 Analog Gain

You can control the gain of pixel signal by analog gain control register. Also, you can control the analog gain of exposure frame individually such as long, middle, and short. As a result, use the following equation to control the analog gain.

Analog Gain
Coarse step x2, Fine step: x1/32 (@ coarse gain 2^n ~ 2^n+1, n = 0, 1, 2, 3)

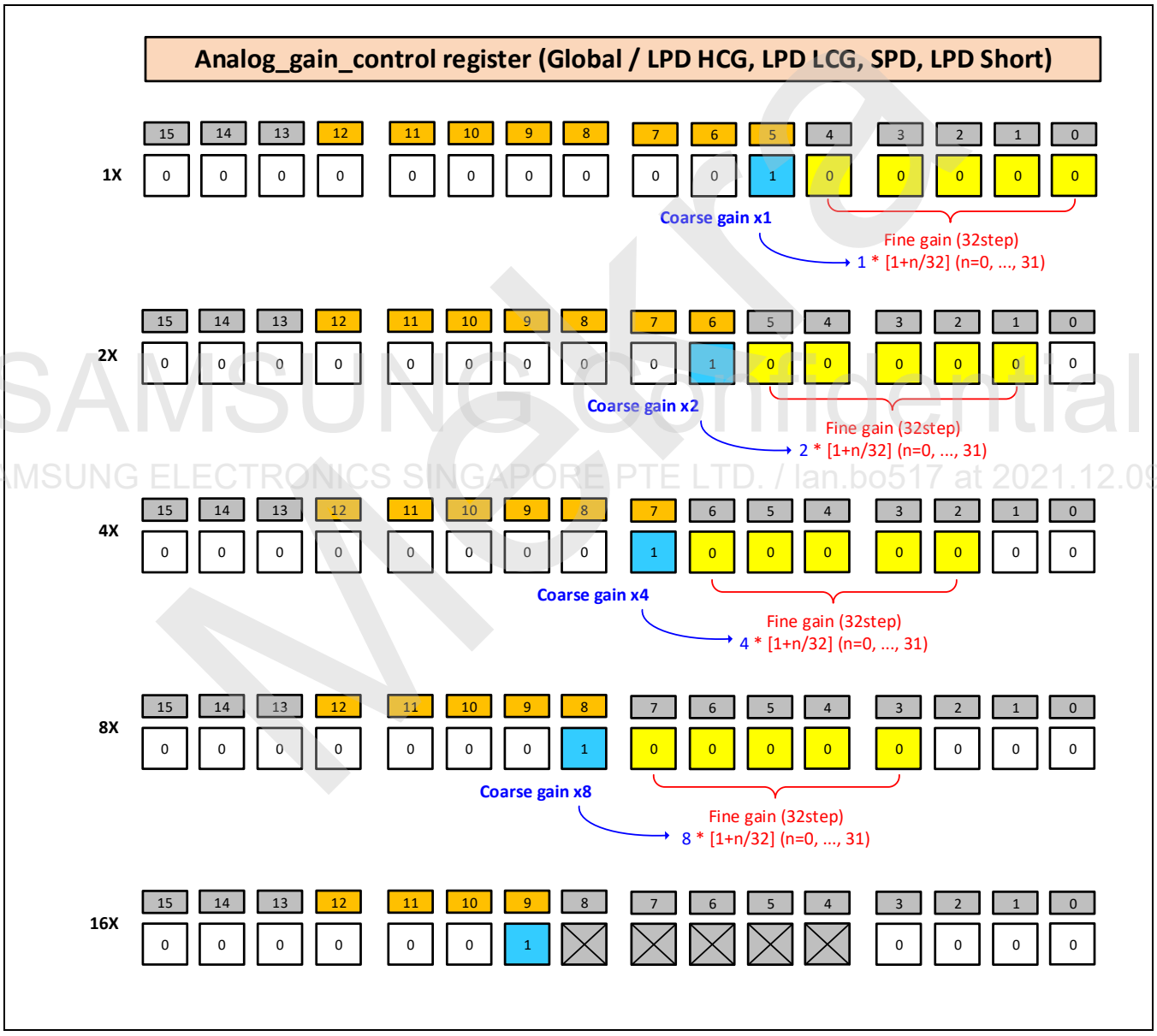


Figure 26 Example of Analog Gain Control

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6.9.2 Digital Gain

You can control the gain of pixel signal for each channel or each exposure type by digital gain control register. As a result, use the following equation to control the digital gain.

- Each exposure type
Digital gain = exposure_gain[15:0]/256
- Each channel
Digital gain = channel_gain[11:0]/4096

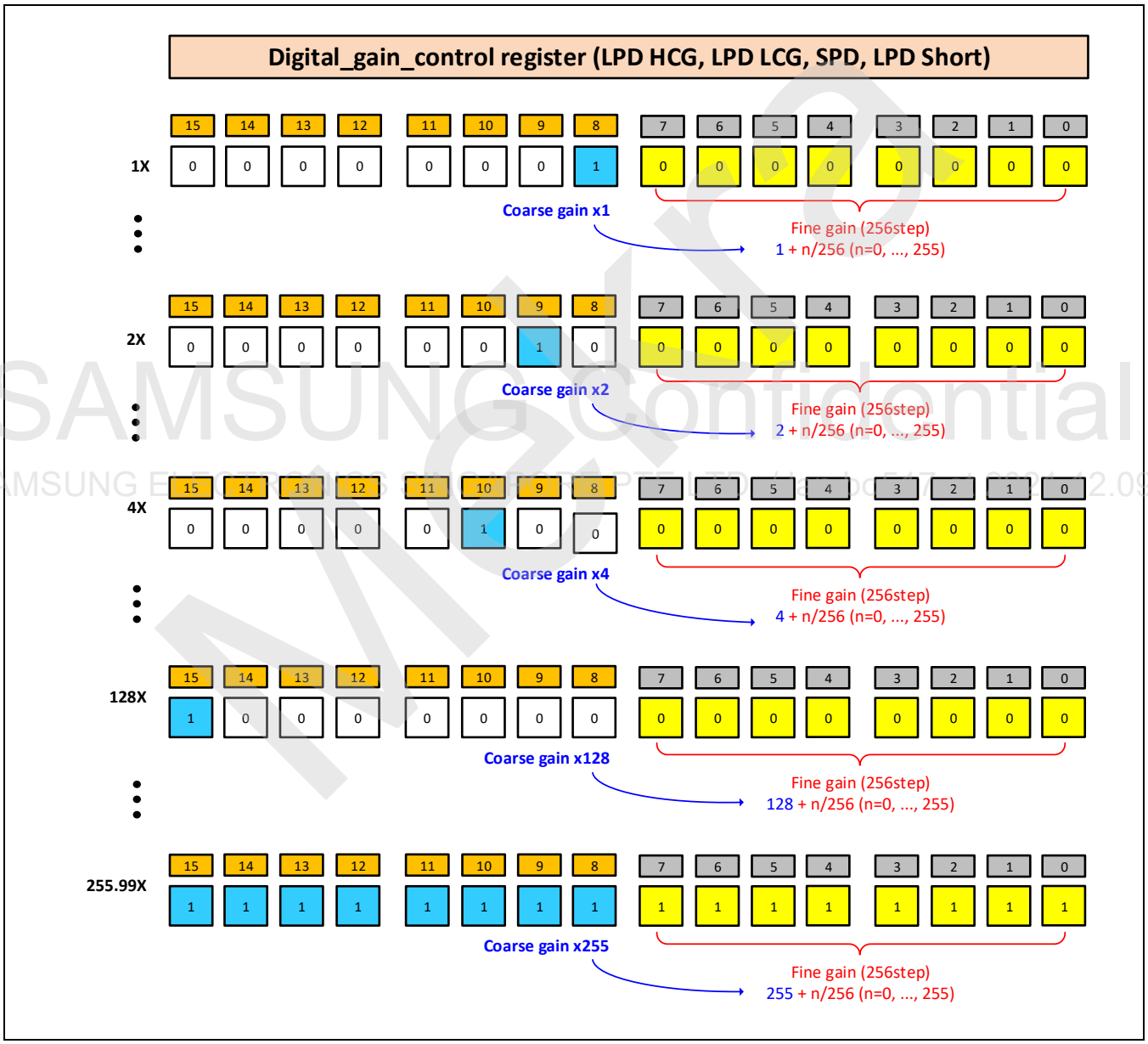


Figure 27 Example of Digital Gain Control (Exposure Type)

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6.10 Auto Dark Level Correction

Auto Dark Level Correction (ADLC) is responsible for row noise reduction and frame black level recovery by subtracting a weighted sum of optical black pixels.

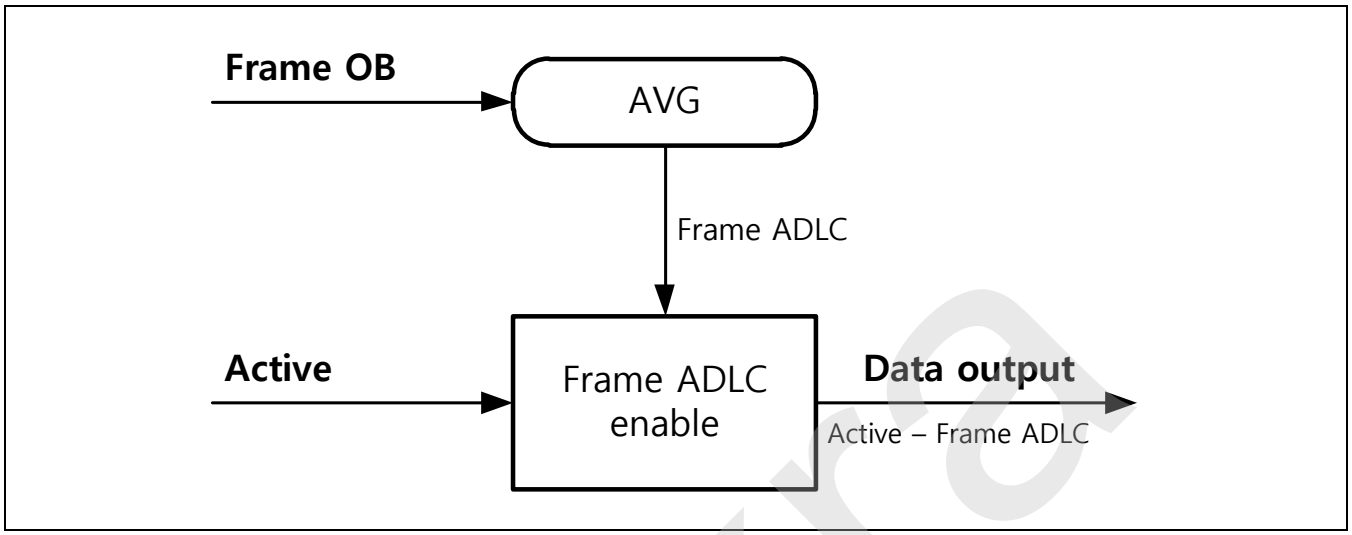


Figure 28 ADLC Algorithm

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6.11 Image Output Format

6.11.1 MIPI

S5K3B6SX MIPI CSI-2 interface is a four-lane high-speed serial interface to connect the camera sensor to a host processor. S5K3B6SX MIPI core IP is compatible with the MIPI alliance standards for D-PHY and CSI2 v2.1.

The maximum high-speed clock frequency of the MIPI is 4.5 GHz.

MIP supports following features:

- MIPI DPHY CSI-2: 4-lane (2/3/4-lane modes) with maximum of 4.5 Gbps (bit clock rate)
- RAW8, RAW10, RAW12, RAW14, RAW16, RAW20 data types
- MIPI transmitting (Tx) only
- One clock lane
- Upto 8 number of virtual channels for the multiple exposure mode
- Ultra-Low-Power Mode (ULPM)

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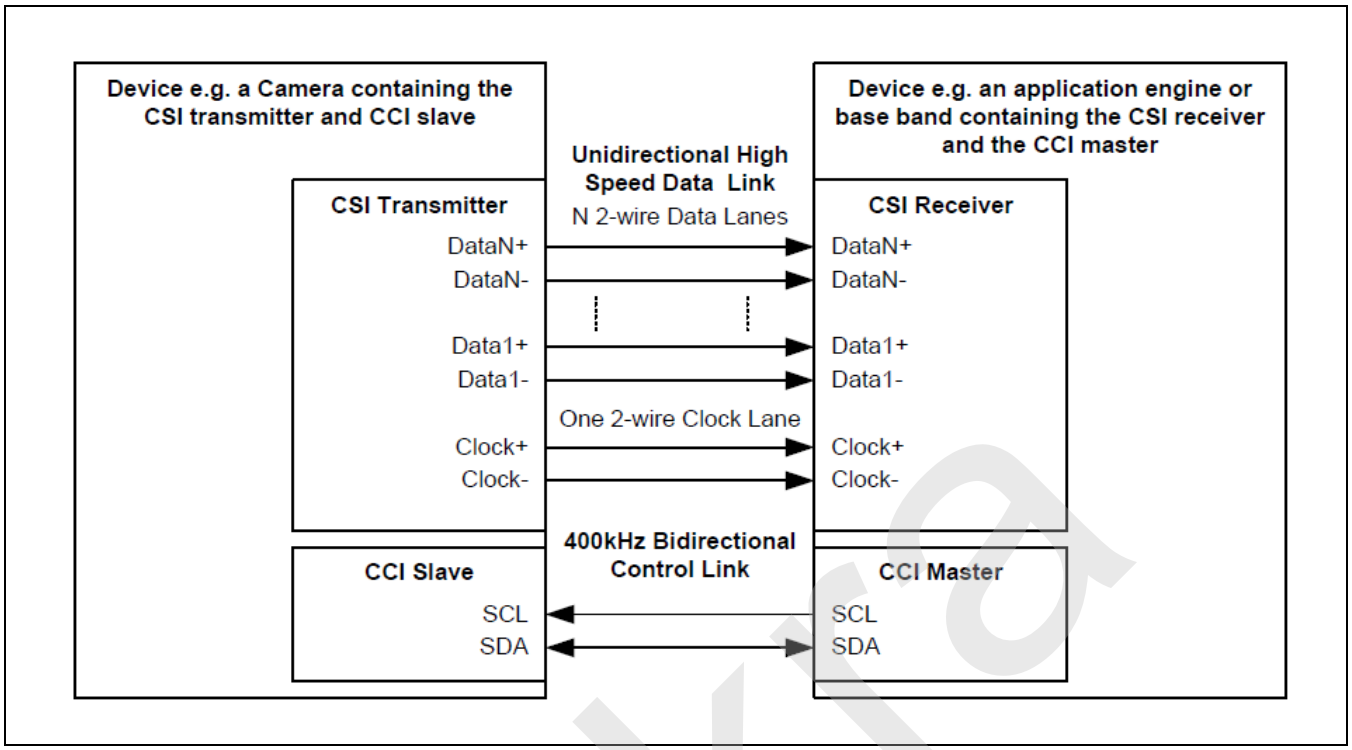


Figure 29 CSI-2 and CCI Transmitter and Receiver Interface (Reference from MIPI CSI-2 v2.1)

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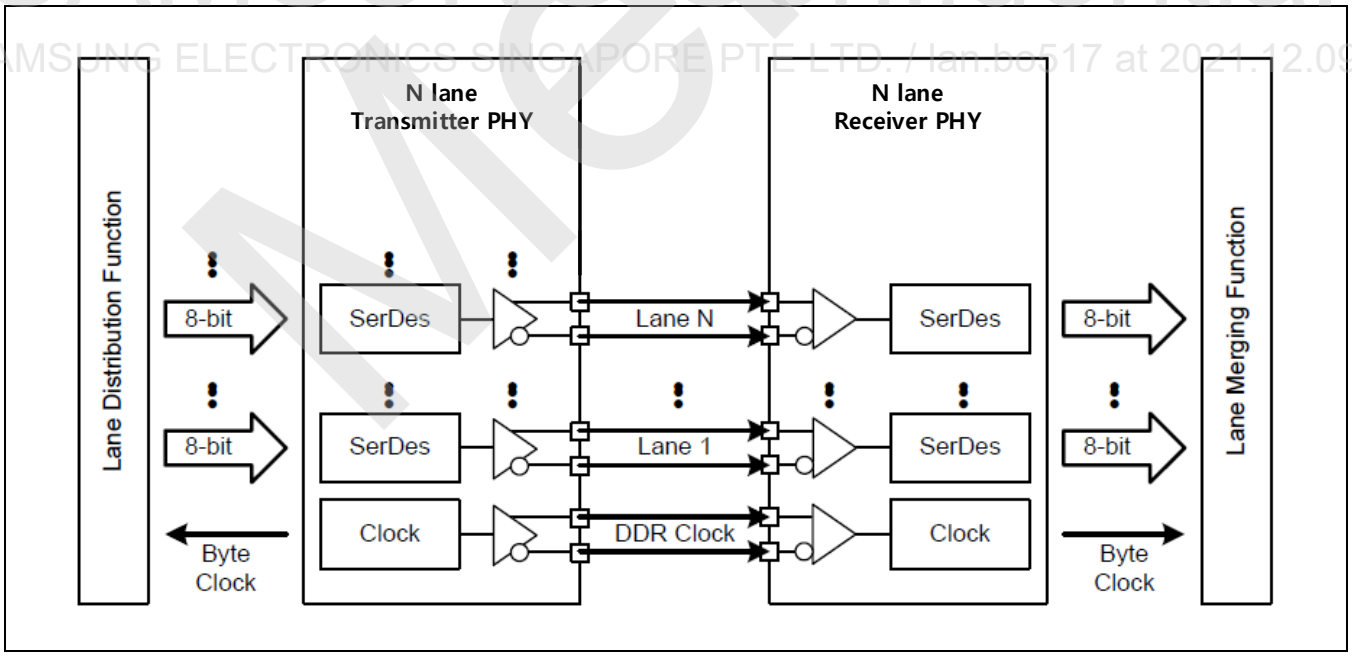


Figure 30 Example of N-Lane Transmitter and N-Lane Receiver (Reference from MIPI CSI-2 v2.1)

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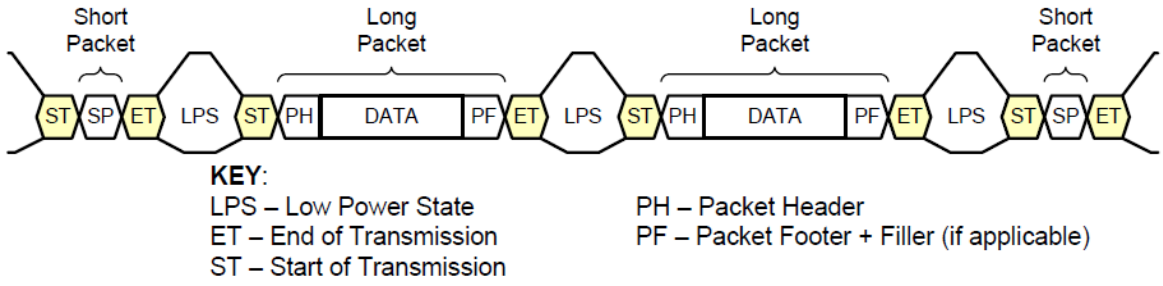


Figure 31 Low-Level Protocol Packet Overview (Reference from MIPI CSI-2 v2.1)

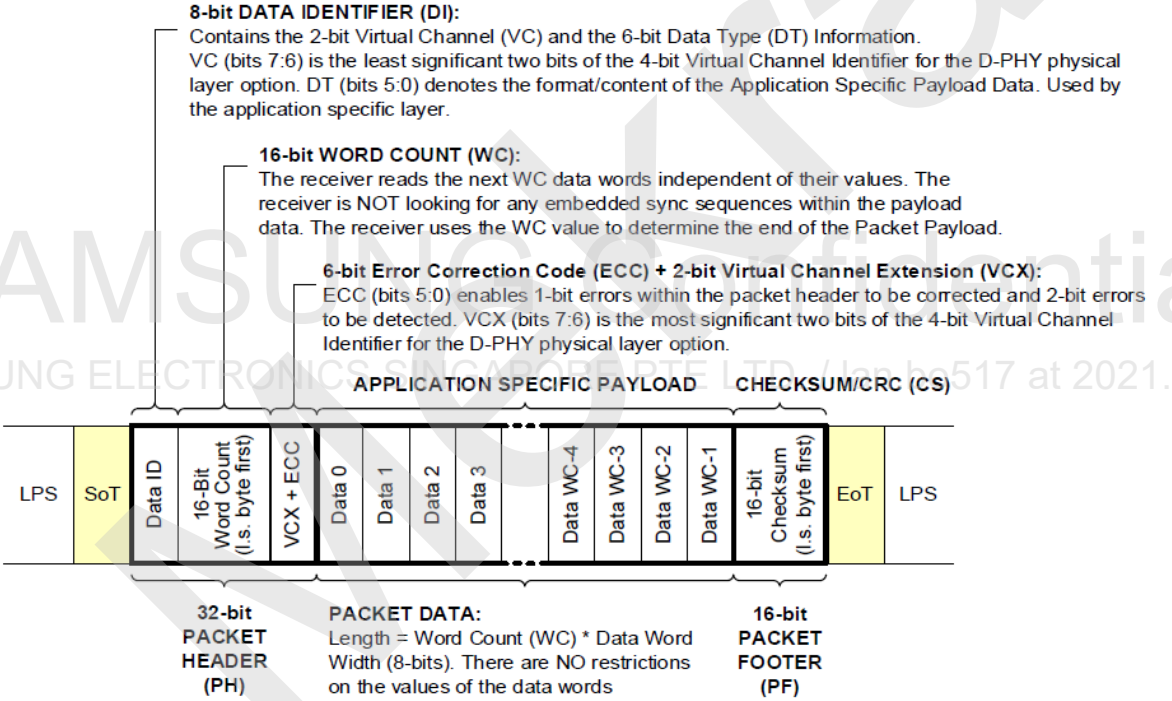


Figure 32 Long Packet Structure (Reference from MIPI CSI-2 v2.1)

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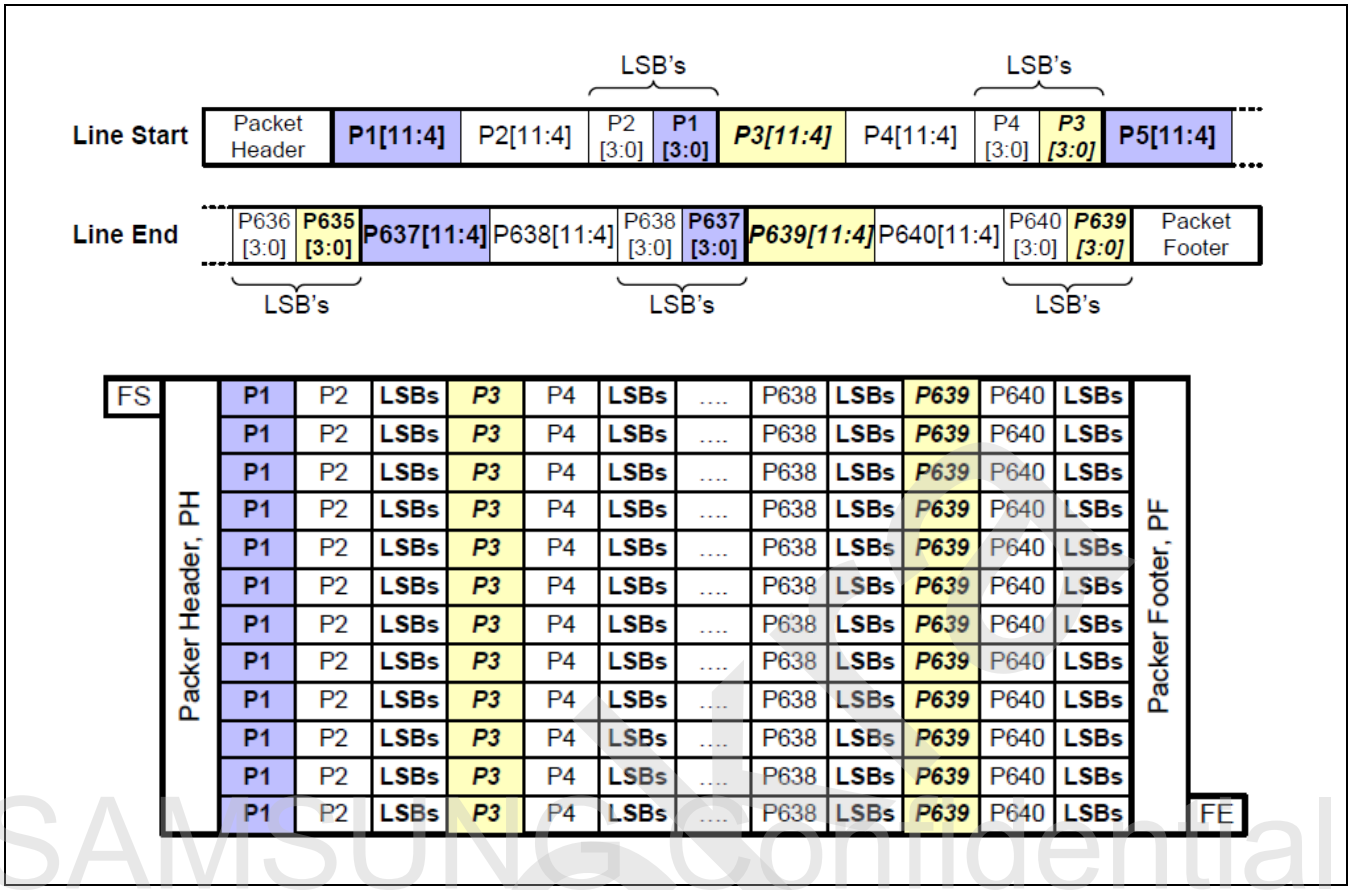


Figure 33 RAW12 Transmission and Frame Format (Reference from MIPI CSI-2 v2.1)

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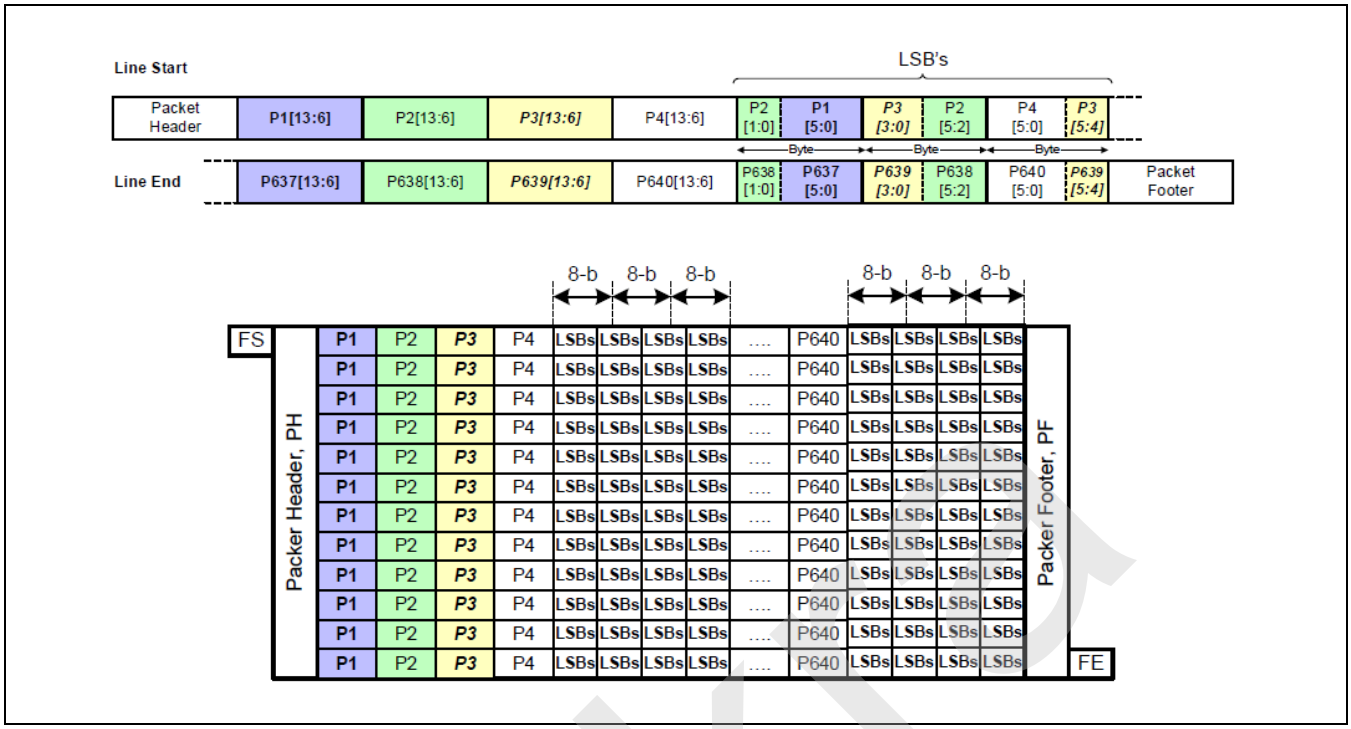


Figure 34 RAW14 Transmission and Frame Format (Reference from MIPI CSI-2 v2.1)

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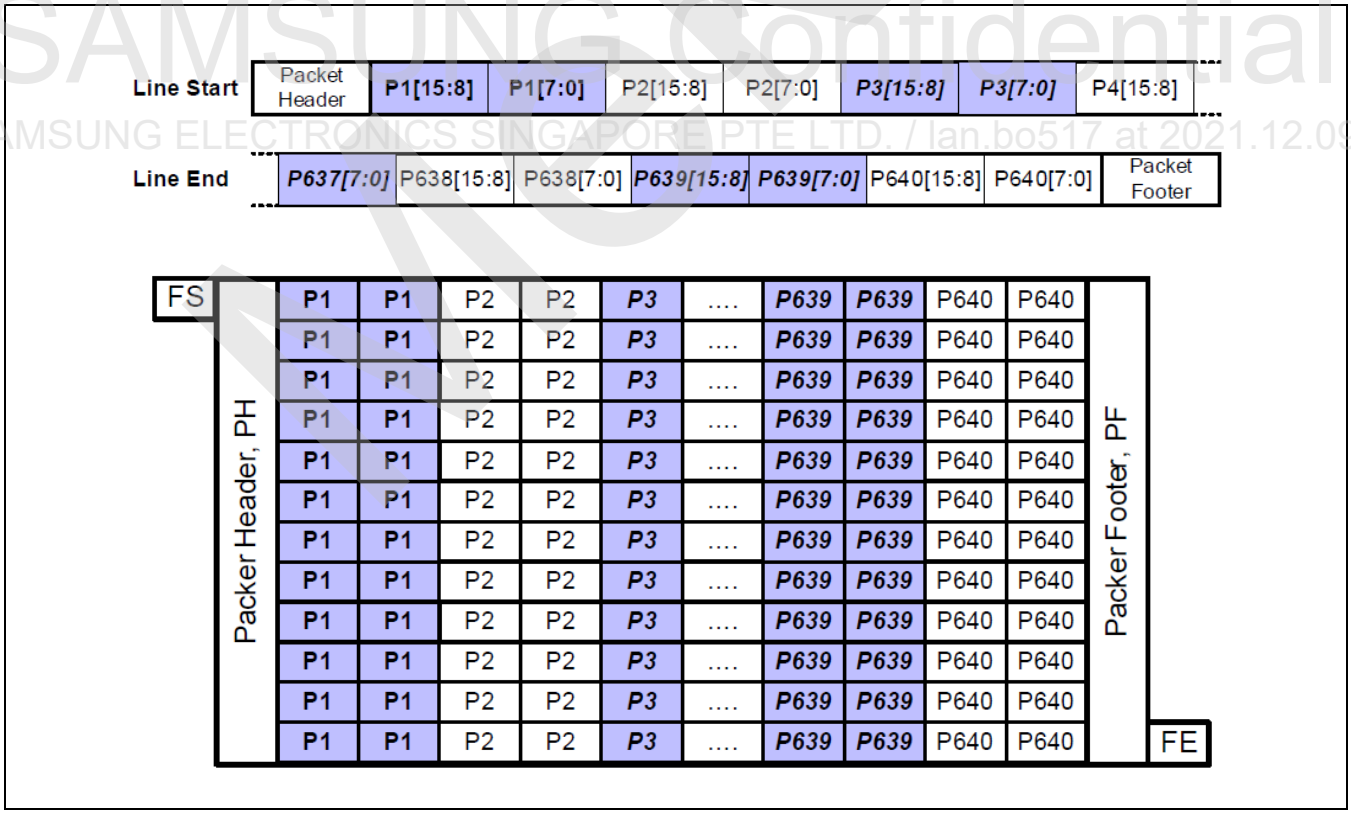


Figure 35 RAW16 Transmission and Frame Format (Reference from MIPI CSI-2 v2.1)

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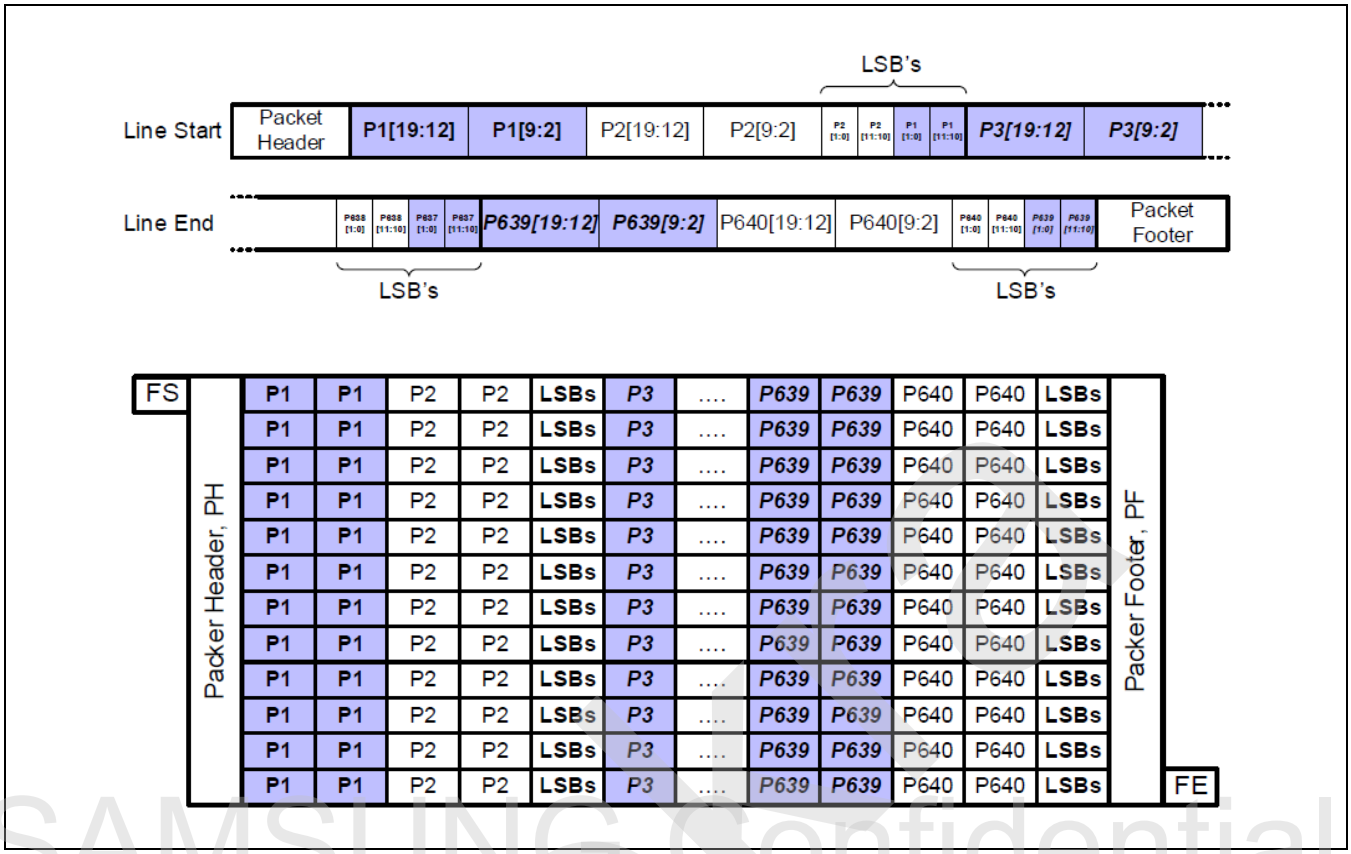


Figure 36 RAW20 Transmission and Frame Format (Reference from MIPI CSI-2 v2.1)

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6.11.1.1 MIPI Virtual Channel Structure

S5K3B6SX able to output various types of data such as Embedded header and footer, Main data which is combined and compressed, Raw and defect and detection map for the reference of back-end processor. To support this, S5K3B6SX has availability to apply virtual channel to those various types of data.

Below table list up various types of data which able to output within same frame and possibility of applying virtual channel at each data when sensor operates intra-scene DCG without LFM mode.

Table 12 Examples of Applying Virtual channel at each data at intra-scene DCG without LFM mode.

Data Type	Output Mode	Data1		Data2		Data3		Data4	
		Data type	VCI	Data type	VCI	Data type	VCI	Data type	VCI
Embedded Header		0x12	O	-					
Main Frame	Line interleaving	RAW12	O	RAW12	O	RAW12	O	RAW12	O
	Pixel interleaving	RAW12	X	-					
	Partially Combined	RAW20	O	-		RAW12	O	Map	O
		RAW16	O	-		RAW12	O	Map	O
	Combined	RAW24	O	-				Map	O
	Compress (PWL)	RAW20	O	-				Map	O
		RAW16	O	-				Map	O
		RAW14	O	-				Map	O
		RAW12	O	-				Map	O
Embedded Footer		0x30	O	-					
Statistics		0x31	O	-					

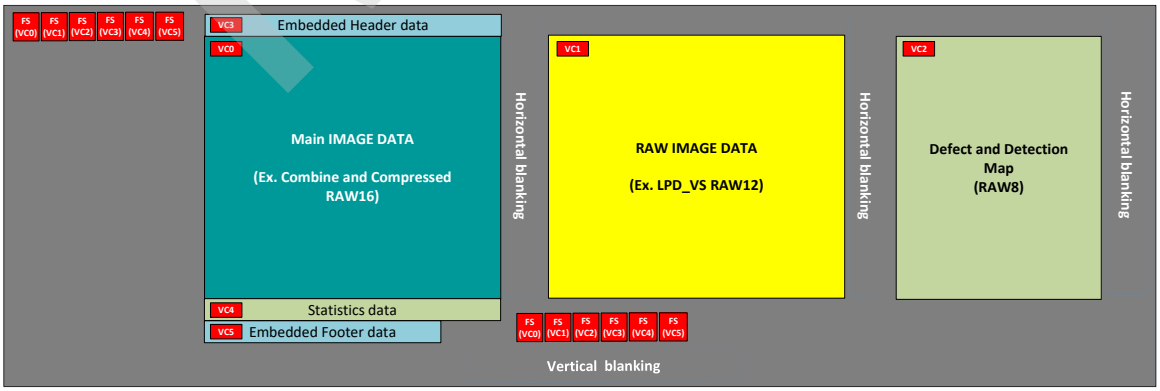


Figure 37 MIPI Virtual Channel for Triple Exposure

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6.11.2 ULPS Mode

S5K3B6SX MIPI Lanes will enter ULPS mode automatically at every software standby mode. DPHY can reduce power consumption by entering ULPS mode.

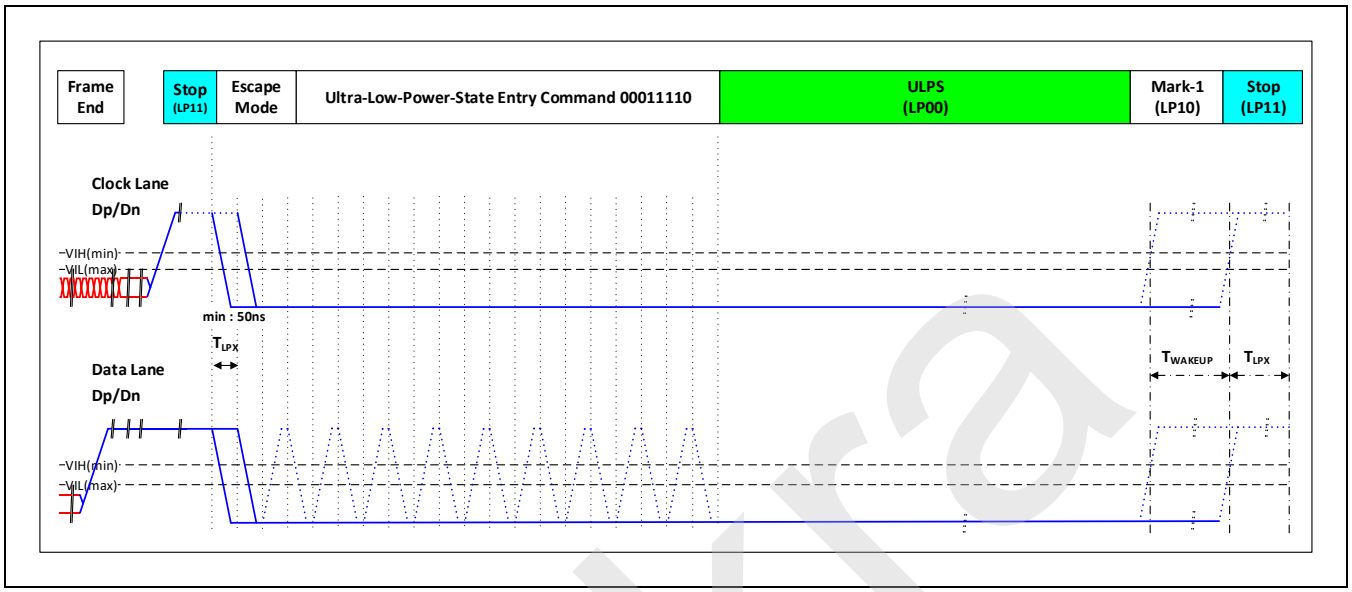


Figure 38 DPHY Ultra-Low-Power State Sequence

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7 Embedded Data and Safety Features

This chapter includes the following sections:

- Embedded data (Header and Footer)
- Safety features for ASIL-B systems

7.1 Embedded Data

Useful embedded data outputs at 0~2 lines on a long exposure frame based on CSI-2 (RAW8 packed data) specification.

Table 13 List of Output Data

Items	Contents
Embedded Header	
Frame information	Horizontal and vertical size, Frame counter
Image size	Horizontal and vertical start and number of readout columns and lines.
Image orientation	Mirror and flip
Sub-sampling ratio	Horizontal and vertical even and odd increments
Register value for AE	Integration time, analog gain, and digital gain
Register value for WDR	Number of each 32 X and Y knee point registers. Registers which relate with WDR algorithm.
Optical black pixel data	Average of optical black pixels
Data format information	RAW 8/RAW10/RAW12/RAW14/RAW16/RAW20
Embedded Footer	
ASIL	Result of internal safety features.
Security	Signature value of frame

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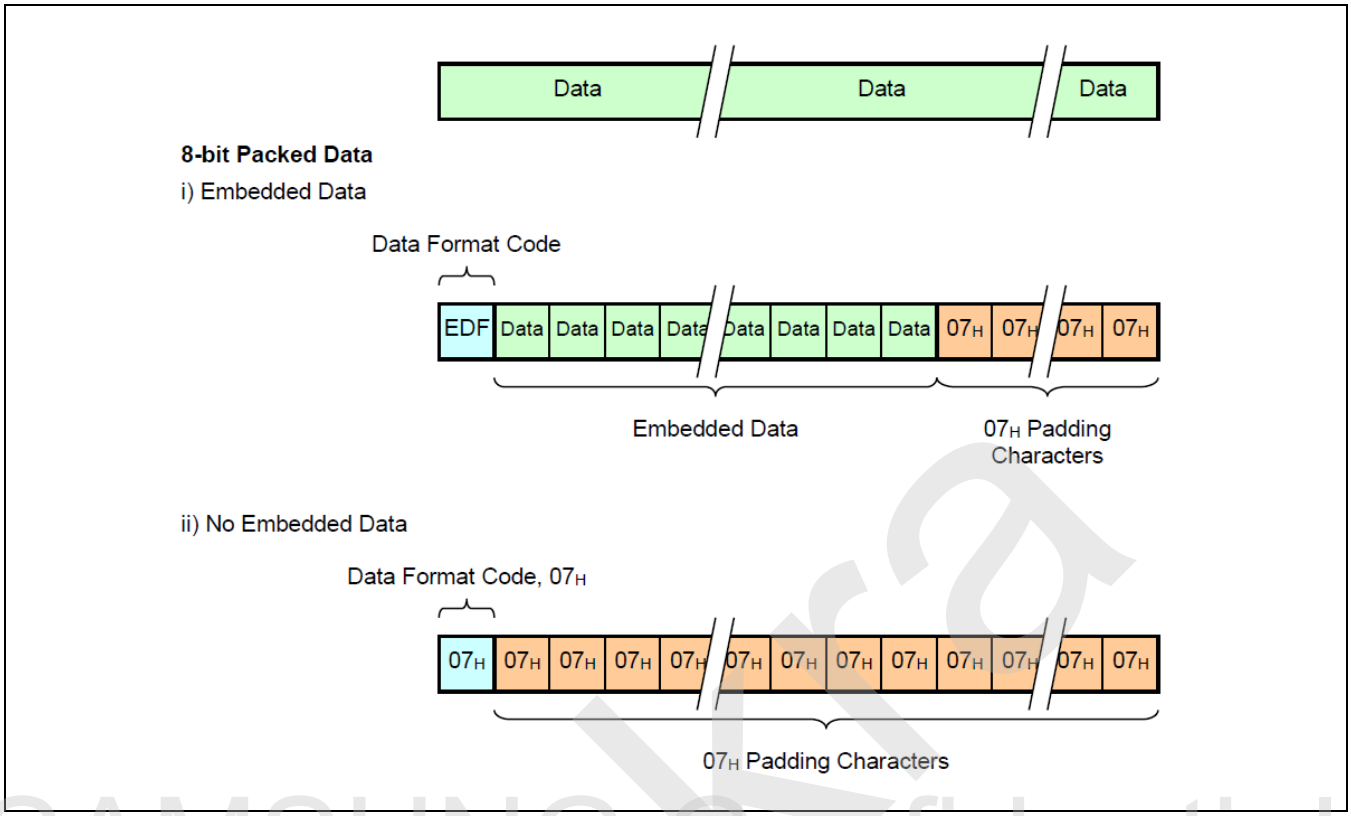


Figure 37 Data Output Stream (Reference from GIIC v0.8)

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7.2 Safety Features for ASIL-B Systems

S5K3B6SX provides safety features for vision systems to meet ASIL-B grade.

Table 14 Features for ASIL-B Systems

Items	Description	Check Period
Memory fault (BIST)	SRAM read-back test detects the stuck at 1 or 0 faults.	EDS (Wafer test)
Memory fault (POST)	SRAM read-back test detects the stuck at 1 or 0 faults when every 'RSTN' release from external.	Every boot-up sequence
Memory ECC	All SRAM include ECC for 1bit error correction and 2bit error detection.	Every frame
Control interface fault	CRC value of CCI data (from host) are put to the embedded data lines. (CRC value is available in case of enable CCI CRC)	User defined
Output interface fault	Image data transmission to ISP or AP contains CRC of data to detect errors in communication.	Every row
Clock Monitoring	Monitoring frequency of clock from System PLL and Interface PLL after Phase Locked.	User defined

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Items	Description	Check Period
E2E Protection	End to End protection at image data and meta data. (Embedded data).	Every frame
Embedded data (Header / Footer)	There are several rows of data which are prepended to the image frame output which are comprised of registers and ASIL supplement data.	Every frame
Digital frame counter	To keep track of the frame count, this value is included as part of the embedded data (64 bit).	Every frame
Test patterns (Digital)	Digital test pattern is pre-pended/appended to each frame of image data.	Every frame /On-demand
Cyber security	Sensor authentication at boot up sequence and secured CRC during data stream.	Every frame
Voltage monitoring	Detect abnormal operation of internal voltage such as stuck to high or low.	During data stream
Temperature sensor	Temperature sensor reports temperature in degrees celsius from -40°C to 125°C within the embedded line data.	Every frame
Pixel control line	This test checks the control signal of pixel transistors.	Every frame
Analog to Digital Converter	This test checks ADC operation of all columns.	Every frame

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8 Electrical Characteristics

This chapter includes the following sections:

- Absolute maximum rating
- Operating conditions
- DC characteristics
- AC characteristics

8.1 Absolute Maximum Rating

Table 15 Absolute Maximum Rating (TBD)

Description	Symbol	Min.	Typ.	Max.	Unit
Digital absolute Max.	VDDD (Max.)	–0.5	–	1.5	V
Analog absolute Max.	VDDA (Max.)	–0.3	–	3.6	
I/O absolute Max.	VDDIO (Max.)	–0.3	–	2.5	
Digital input voltages absolute Max (1)	VIP (Max.)	–0.3	–	VDDIO + 0.3	
VCAP analog voltage absolute Max (2)	VCAP	–0.3	–	4.2	
Storage temperature	TSTR	–40	–	-40 to +125	°C

NOTE:

The values beyond the absolute maximum ratings damage the devices. They do not imply any functional operation of the device at these conditions or any other conditions beyond the recommended operating conditions.

1. Digital Inputs: MCLK, RSTN, SCE, SDA, SCL, GPIO_*, VSYNC_IN, HSYNC_IN
2. Voltage on external analog capacitors

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8.2 Operating Conditions

Table 16 Operating Conditions

Description	Symbol	Min.	Typ.	Max.	Unit
Analog supply voltage	VDDA	2.66	2.8	2.94	V
I/O supply (low voltage)	VDDIO	1.7	1.8	1.9	V
Digital supply	VDDD	0.95	1.05	1.15	V
Operating temperature	TOPR (1)	-40	-	125	°C

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NOTE:

1. TOPR is the junction operating temperature of sensor function.

8.3 DC Characteristics

Table 17 DC Characteristics

Description	Symbol	Min.	Typ.	Max.	Unit
HIGH level input voltage (LVCMOS interface) VDDIO = 1.8 V	VIH	0.7 x VDDIO*	-	-	V
Low level input voltage (LVCMOS interface) VDDIO = 1.8 V	VIL	-	-	0.3 x VDDIO	V
Input leakage current (1)	IIH (Vin=VDDIO) (2)	-	-	15	µA
	IIL (Vin=VSSIO) (2)	-15	-	-	µA
Input capacitance	CIN (3)	-	-	5	pF
Input operating frequency	FIN (MCLK)	6	27	64	MHz
Analog supply operating current (VPIX, VDDA, VDDA_MIPI)	IDDA	-	TBD	-	mA
Digital supply operating current	IDDD	-	TBD	-	mA
I/O supply operating current (low voltage)	IDDIOL	-	TBD	-	mA
I/O supply operation current (high voltage)	IDDIOH	-	TBD	-	mA

NOTE:

1. When the sensor package is soldered on PCB, IIH, and IIL of SDA, SCL pins depend on the value of the off-chip pull-up devices such as resistors. This values are used for low I/O voltage scenarios (VDDIO=1.8 V typ.)
2. Vin means the input voltage of each package pin.
3. CIN max. can be varied based on the environment.

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8.4 AC Characteristics

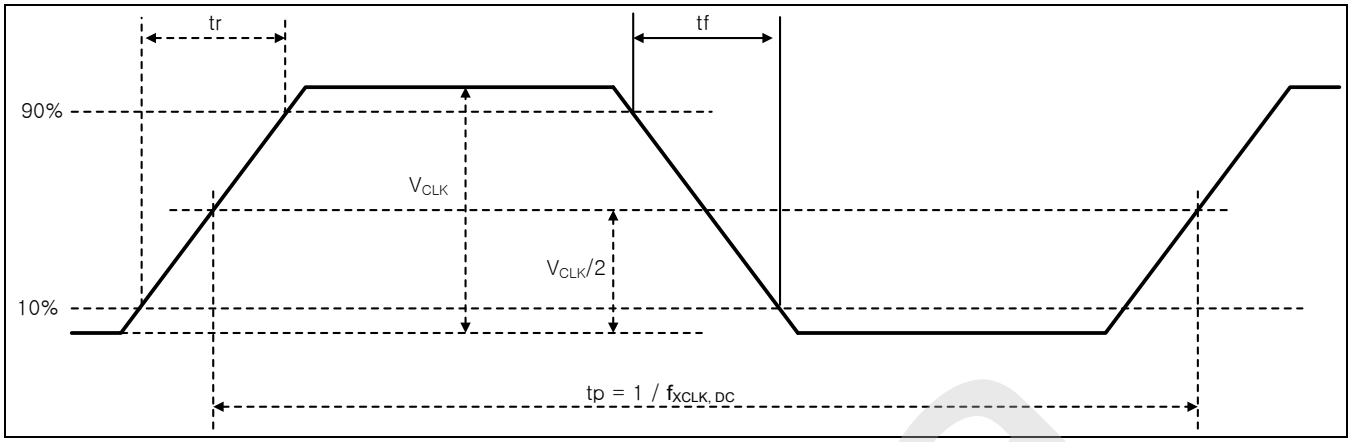


Figure 38 Master Clock Waveform

Table 18 AC Characteristics

Characteristics	Symbol	Condition	Min.	Typ.	Max.	Unit
External clock frequency (NOTE)	$f_{XCLK, DC}$	-	6	-	64	MHz
External clock duty cycle (NOTE)	f_{XDUTY}	-	45	-	55	%
External clock period jitter (peak-to-peak)	T_{jitter}	-	-	-	600	ps
External clock rise and fall times	-	-	-	-	6	ns
PLL locking time	t_{LOCK}	-	-	-	500	µs

NOTE: It is applicable for MCLK pin.

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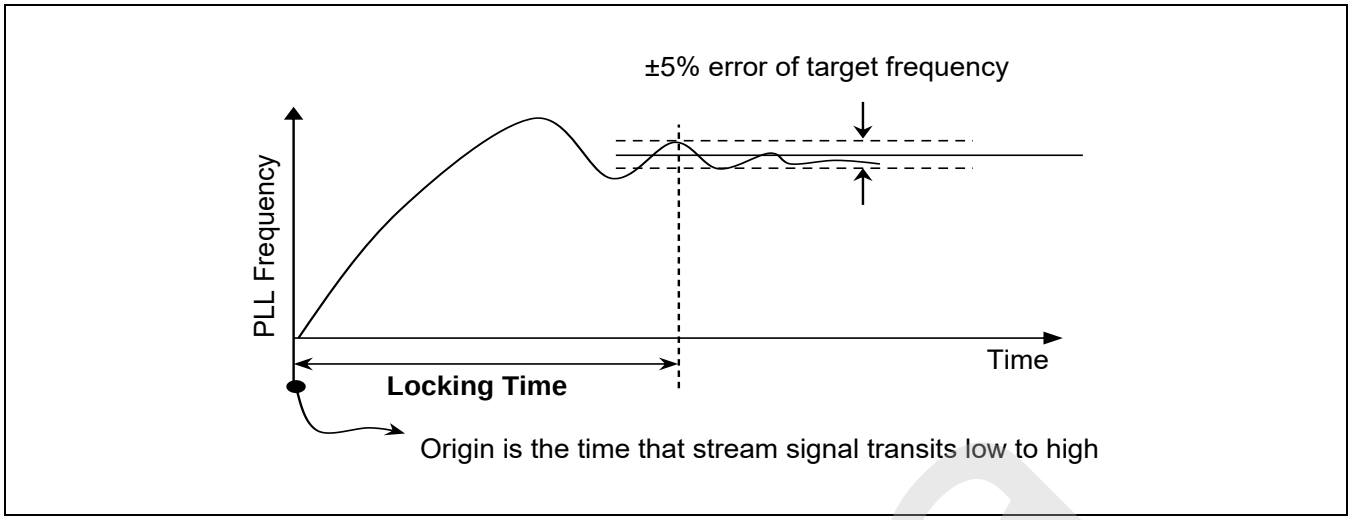


Figure 39 PLL Locking Time

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